Summary of available Northwest Fishery Science Center survey data to support assessments of U.S. West Coast groundfish stocks

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1 Introduction

This document provides a detailed summary of commonly used data sources that may be used to support assessments in 2027. A detailed summary of data available by year and across sources can allow the Pacific Fishery Management Council (Council) and advisory bodies to understand the coverage of data across time and the potential viability of a new assessment or assessment type.

HIGHLIGHT CHANGES TO THIS YEAR'S REPORT

1.1 Biological Samples

Data from the Northwest Fisheries Science Center (NWFSC) West Coast Groundfish Bottom Trawl Survey (WCGBT) and NWFSC Hook-and-Line (HKL) surveys are summarized.

Data collected by the NWFSC WCGBT survey between 2003-2024 and the NWFSC HKL survey from 2004-2024 are summarized by species. Lengths, aged fish, and unread age structures collected are available by year. Additionally, the number of tows (NWFSC WCGBT survey) or sites (NWFSC HKL survey) that observed each species by year are also provided.

1.2 Survey Length Compositions

The length data collected by the NWFSC WCGBT survey were expanded using a generalized area-based stratification. The composition data were expanded using a design-based approach with strata based on state latitudes with two depth strata: 55 - 183 m and 183 - 549 m, with an exception for four species. The four species have considerable biomass at depths greater than 549 m: sablefish, Dover sole, longspine thornyhead, and shortspine thornyhead. Thus an additional depth strata that covered deeper waters, 549 - 1,280 m, for each state area was added. The expanded length composition data were summarized using either a 2 or 4 cm bin structure depending upon the range between maximum and minimum lengths observed within the survey data. Species where the range between the maximum and minimum lengths observed by the survey were less than 60 cm, 2 cm data bins were used, while for species where the range was 60 cm or greater the data bins were set at 4 cm. All length observations were treated as unsexed fish in this exercise for simplicity and for ease of observing potential trends in length observations across time. The generalized stratification and bin structure selected here provides a simple summary of the data that can be useful for decision making, but will likely differ from a species-specific approach that would be selected in a future assessment. Additionally, the NWFSC WCGBT survey selectivity for each species will impact the lengths observed and has not been explicitly accounted for in this analysis.

The length data collected by the NWFSC HKL survey were summarized to reflect the proportion of observations by species, length bin, and year. The length composition data were summarized using either a 2 or 4 cm bin structure depending upon the range between maximum and minimum lengths observed within the survey data, in the same manner as for the NWFSC WCGBT survey. Similar to the NWFSC WCGBT survey, the selectivity of the NWFSC HKL survey for each species will impact the lengths observed and has not been explicitly accounted for in this analysis.

1.3 Survey Relative Indices of Abundance

Indices of abundance were estimated from the NWFSC WCGBT survey using a spatiotemporal model (smdTMB) for species well-observed by the survey. The indices were estimated using a generalized set-up across species using either a gamma or log-normal distribution, with the dispersion parameter held constant across years with 200-500 knots used to create the mesh. The model structure for each species is provided in Section 3. Future species-specific indices of abundance created with sdmTMB would likely select a more tailored approach for model settings (e.g., knots, distributions) which could result in slight changes in year-by-year values for the indices of abundance. The indices of abundance presented here should only be considered illustrative of potential trends in abundance across time.

Indices of abundance were estimated from the NWFSC HKL survey for well-observed species using a negative-binomial model that accounted for year, site, and drop number. Additional covariates that would typically be explored when developing species-specific indices for use in stock assessments were not explored in this analysis for simplicity. The selection of how to model these data for a particular species may vary from the approach applied here. For example, the recent vermilion and sunset rockfish assessment modeled the data as two indices (i.e., one for areas outside of the Cowcod Conservation Areas [CCA] and one for sites inside the CCAs). In contrast, the recent assessment of copper rockfish estimated a single index with observations by areas weighted (nearshore sites, northern Channel Islands, Southern Channel Islands).

1.4 Maturity Data Collections

Maturity samples for a wide range of West Coast groundfish species have been collected across a range of sources: NWFSC WCGBT survey, NWFSC HKL survey, Pacific hake survey, at-sea sampling of the Pacific hake fishery, and port sampling by ODFW and WDFW. Samples have been collected between 2009 - 2024. The following summary does not include collection from the 2025 NWFSC WCGBT and HKL surveys. Summaries of the maturity collections is provided for each species. This data summary only includes collections led by the NWFSC but other collections could be available at the SWFSC, CDFW, ODFW, WDFW, or other research groups. Future data summaries will look to integrate these additional collections in subsequent versions of this document.

1.5 Additional Data

Data may be available for consideration in future assessments that are currently not included in this report. A summary of potential additional data that could be available are described below:

- PacFIN and RecFIN.

- Emphasize again that this is only survey data.

2 Groundfish species observed by surveys conducted by the NWFSC

2.1 Arrowtooth flounder

The most recent assessment of arrowtooth flounder was an update assessment conducted in 2017. Across available data, arrowtooth flounder have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 215 positive tows per year. Across the coast a total of 254 maturity samples have been collected with 96 read maturities.

Table 1: Total number of available lengths, read ages, and unread age structures by data source and state between for arrowtooth flounder.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	9,814	802	2,772
OR	NWFSC WCGBTS	29,433	2,245	7,046
WA	NWFSC WCGBTS	19,722	1,281	3,896

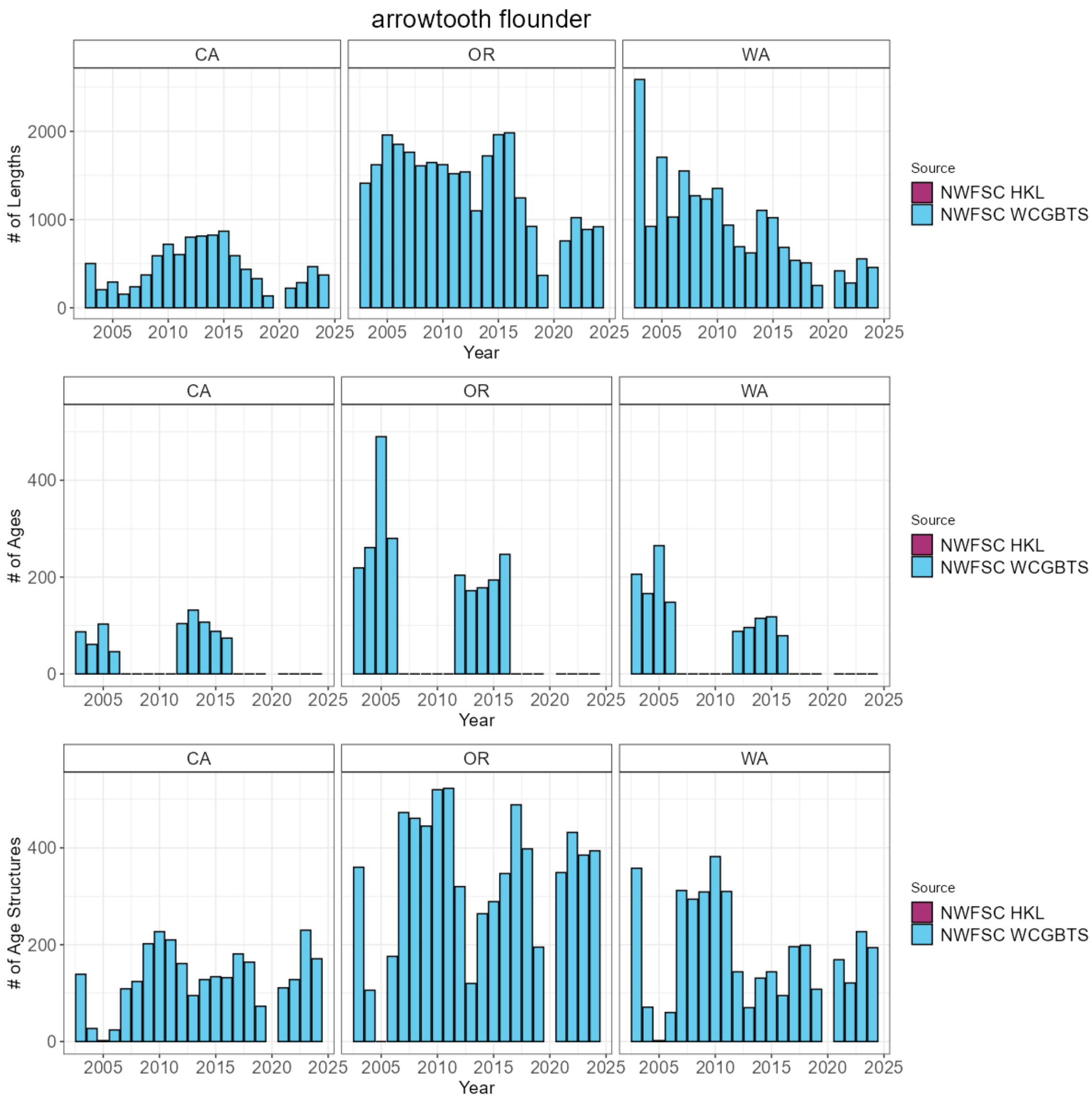

Figure 1: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.2 Aurora rockfish

The most recent assessment of aurora rockfish was a benchmark assessment conducted in 2013. Across available data, aurora rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 80 positive tows per year. Across the coast a total of 617 maturity samples have been collected with 567 read maturities.

Table 2: Total number of available lengths, read ages, and unread age structures by data source and state between for aurora rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	26,219	2,269	7,988
OR	NWFSC WCGBTS	5,453	784	2,843
WA	NWFSC WCGBTS	384	37	258

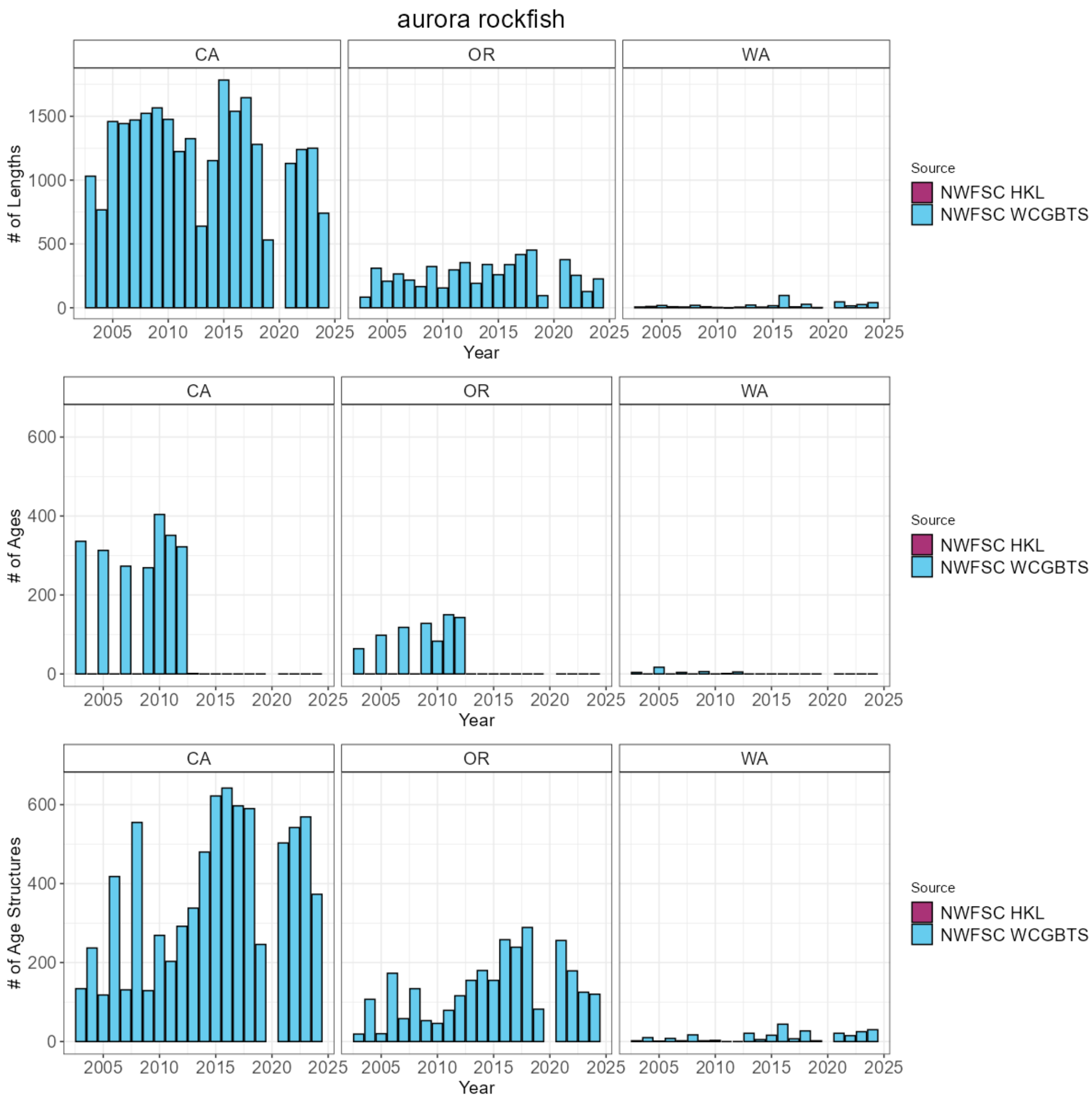

Figure 2: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.3 Bank rockfish

To date, no assessment or analysis has been conducted on bank rockfish. Across available data, bank rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 27 positive tows per year. The NWFSC WCGBTS survey has an average of 12 positive tows per year. Across the coast a total of 733 maturity samples have been collected with 85 read maturities.

Table 3: Total number of available lengths, read ages, and unread age structures by data source and state between for bank rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	3,666	0	3,651
CA	NWFSC WCGBTS	2,097	0	1,455
OR	NWFSC WCGBTS	142	0	55
WA	NWFSC WCGBTS	4	0	4

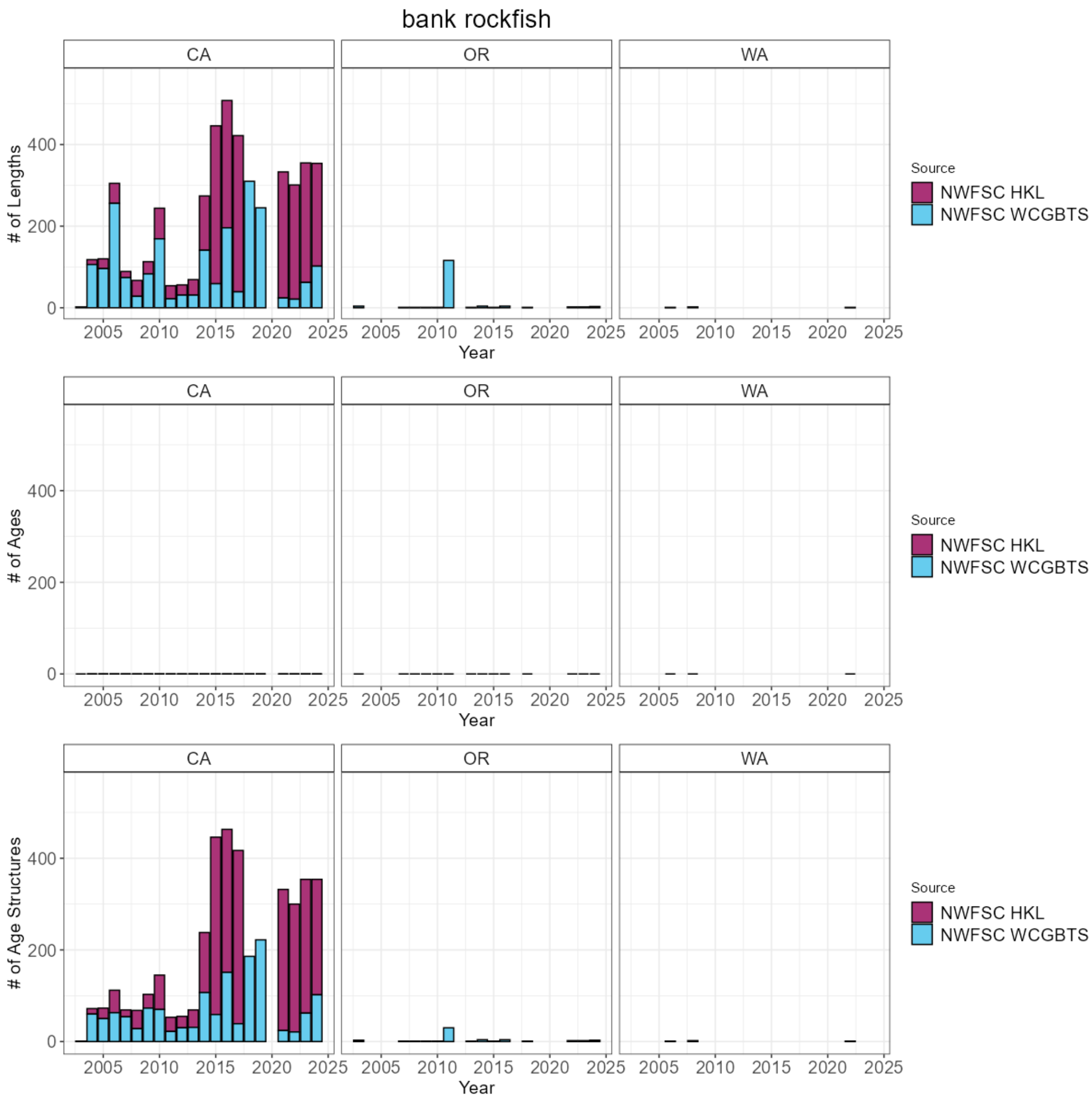

Figure 3: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.4 Big skate

The most recent assessment of big skate was a benchmark assessment conducted in 2019. Across available data, big skate have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 92 positive tows per year. Across the coast a total of 180 maturity samples have been collected with 180 read maturities.

Table 4: Total number of available lengths, read ages, and unread age structures by data source and state between for big skate.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	2,263	351	132
OR	NWFSC WCGBTS	2,651	422	331
WA	NWFSC WCGBTS	1,987	261	235

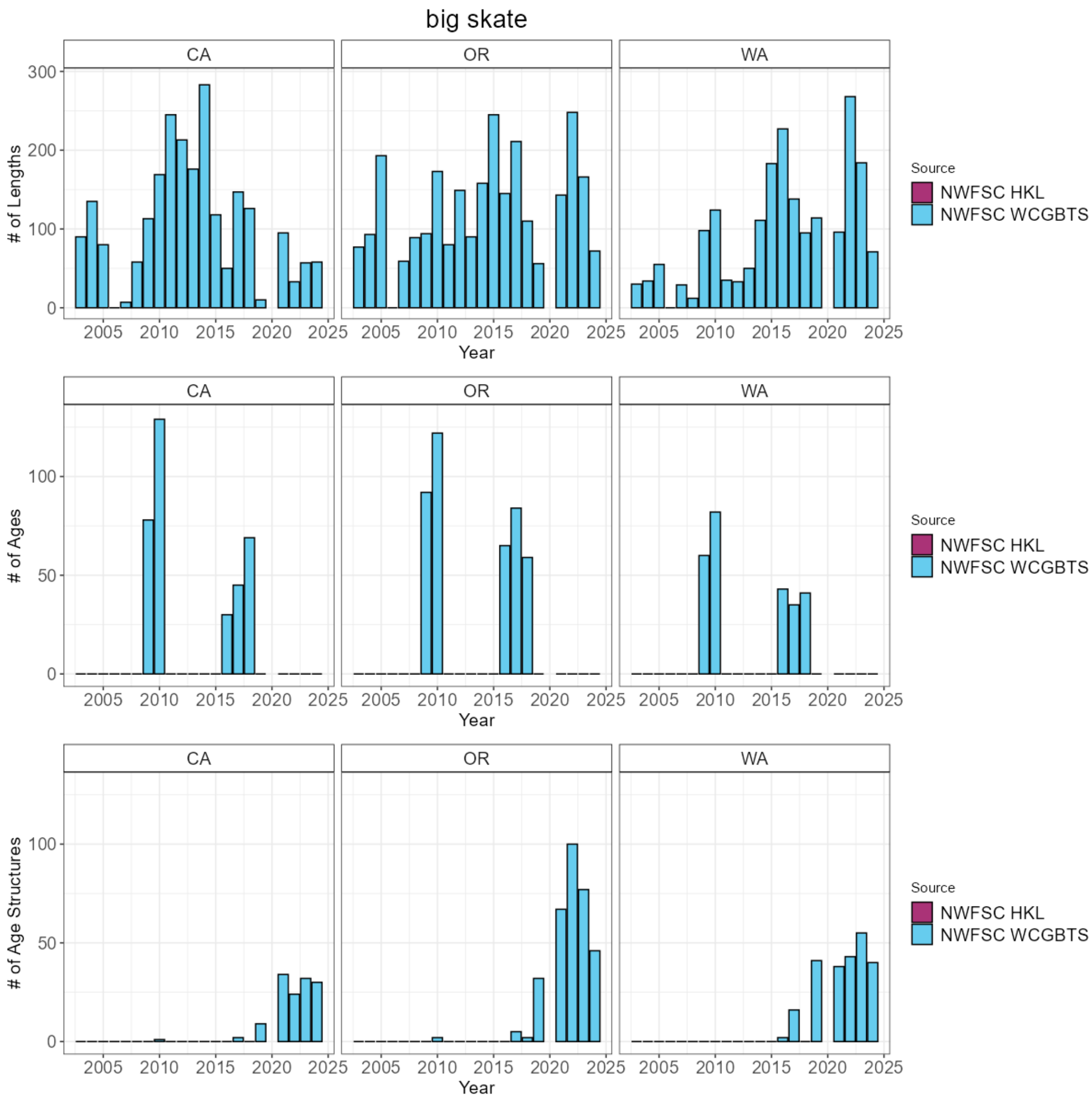

Figure 4: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.5 Blackgill rockfish

The most recent assessment of blackgill rockfish was an update assessment conducted in 2017. Across available data, blackgill rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 34 positive tows per year. Across the coast a total of 126 maturity samples have been collected with 126 read maturities.

Table 5: Total number of available lengths, read ages, and unread age structures by data source and state between for blackgill rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	7	0	7
CA	NWFSC WCGBTS	10,484	1,937	5,586
OR	NWFSC WCGBTS	286	11	254
WA	NWFSC WCGBTS	7	0	7

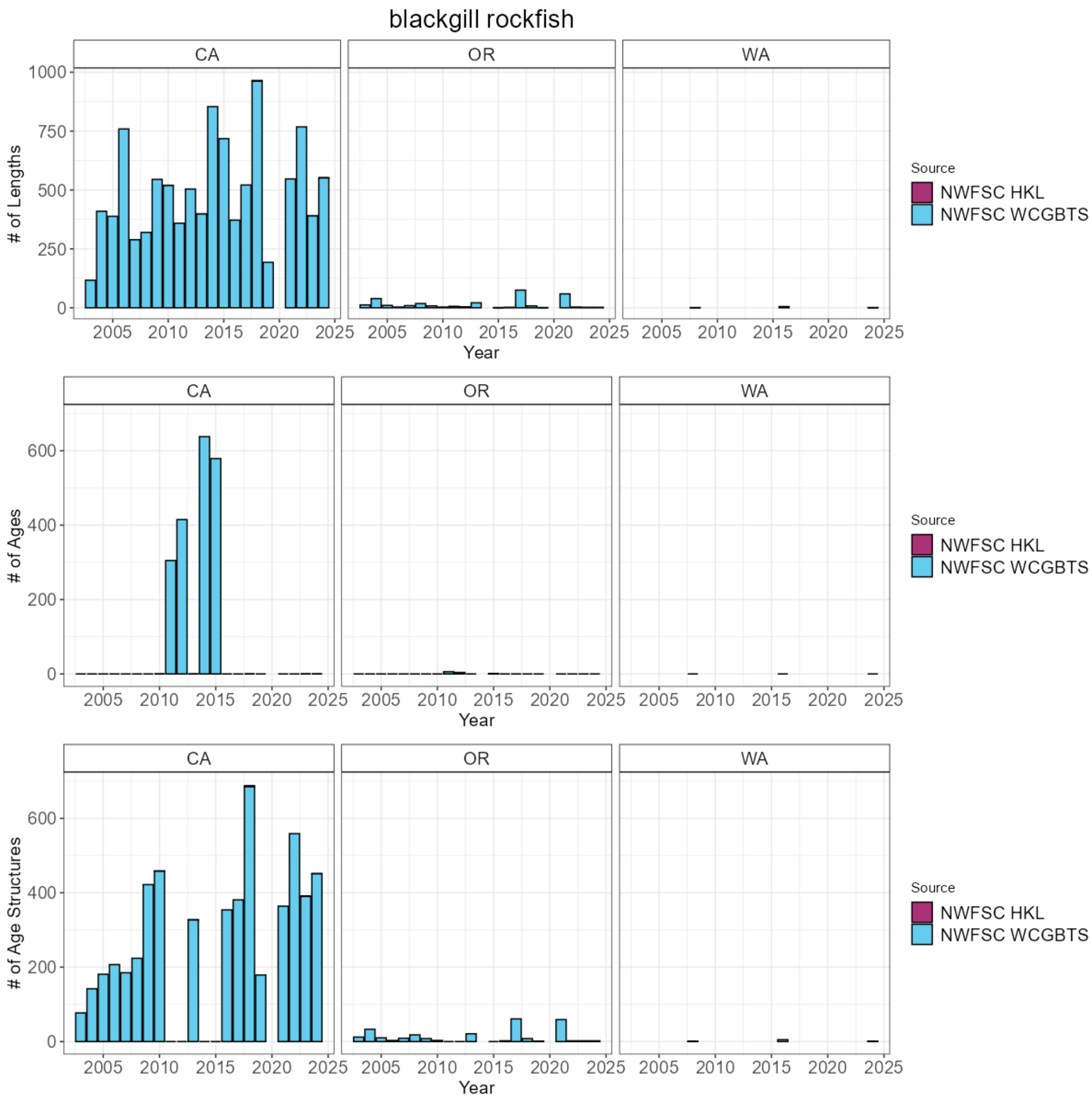

Figure 5: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.6 Bocaccio

The most recent assessment of bocaccio was an update assessment conducted in 2017. Across available data, bocaccio have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 112 positive tows per year. The NWFSC WCGBTS survey has an average of 49 positive tows per year. Across the coast a total of 837 maturity samples have been collected with 737 read maturities.

Table 6: Total number of available lengths, read ages, and unread age structures by data source and state between for bocaccio.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	21,333	0	13,842
CA	NWFSC WCGBTS	10,436	2,759	3,969
OR	NWFSC WCGBTS	294	20	180
WA	NWFSC WCGBTS	477	74	333

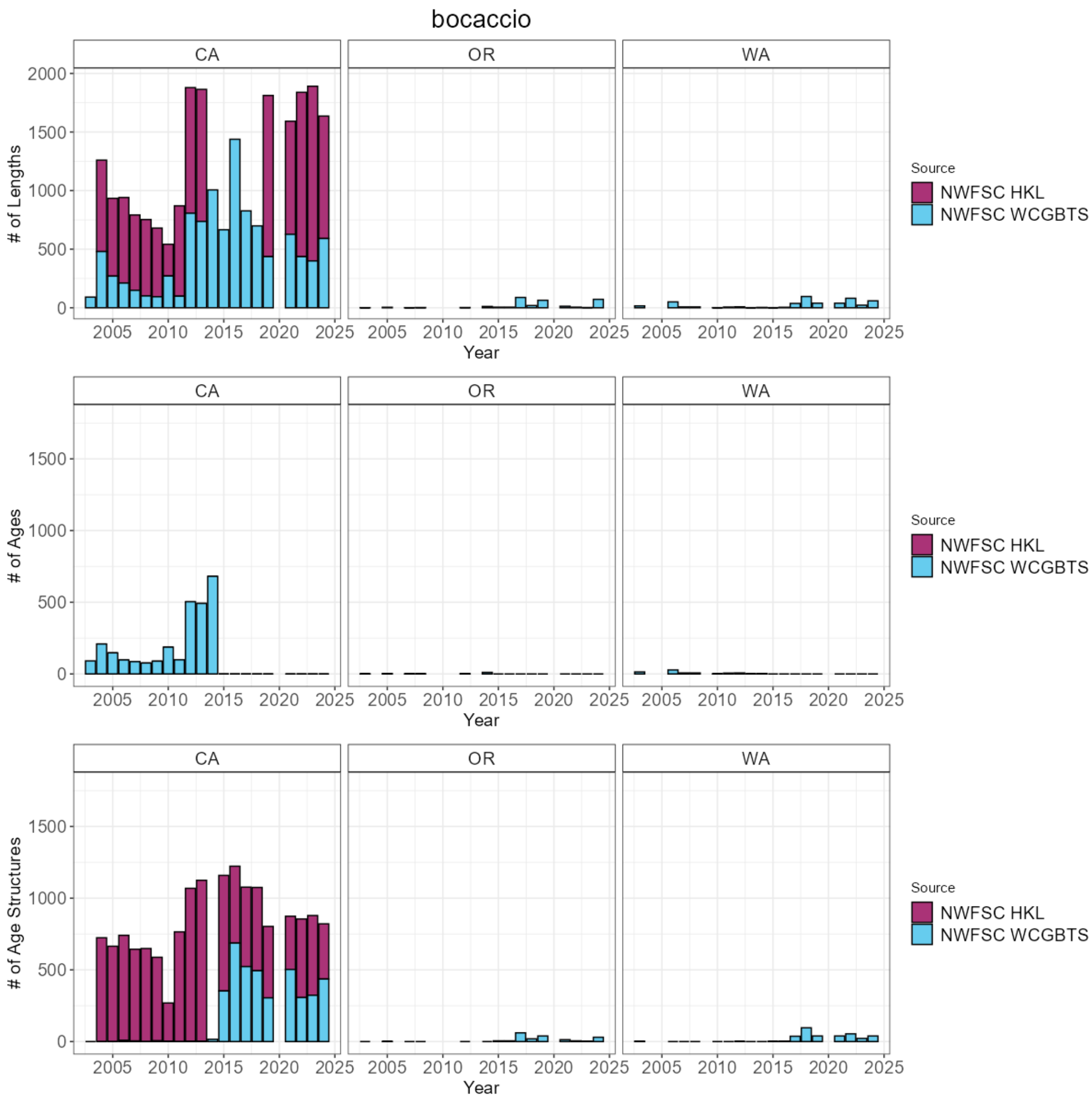

Figure 6: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.7 California scorpionfish

The most recent assessment of California scorpionfish was a benchmark assessment conducted in 2017. Across available data, California scorpionfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC WCGBTS survey has an average of 12 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 7: Total number of available lengths, read ages, and unread age structures by data source and state between for California scorpionfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	41	0	23
CA	NWFSC WCGBTS	4,203	911	1,044

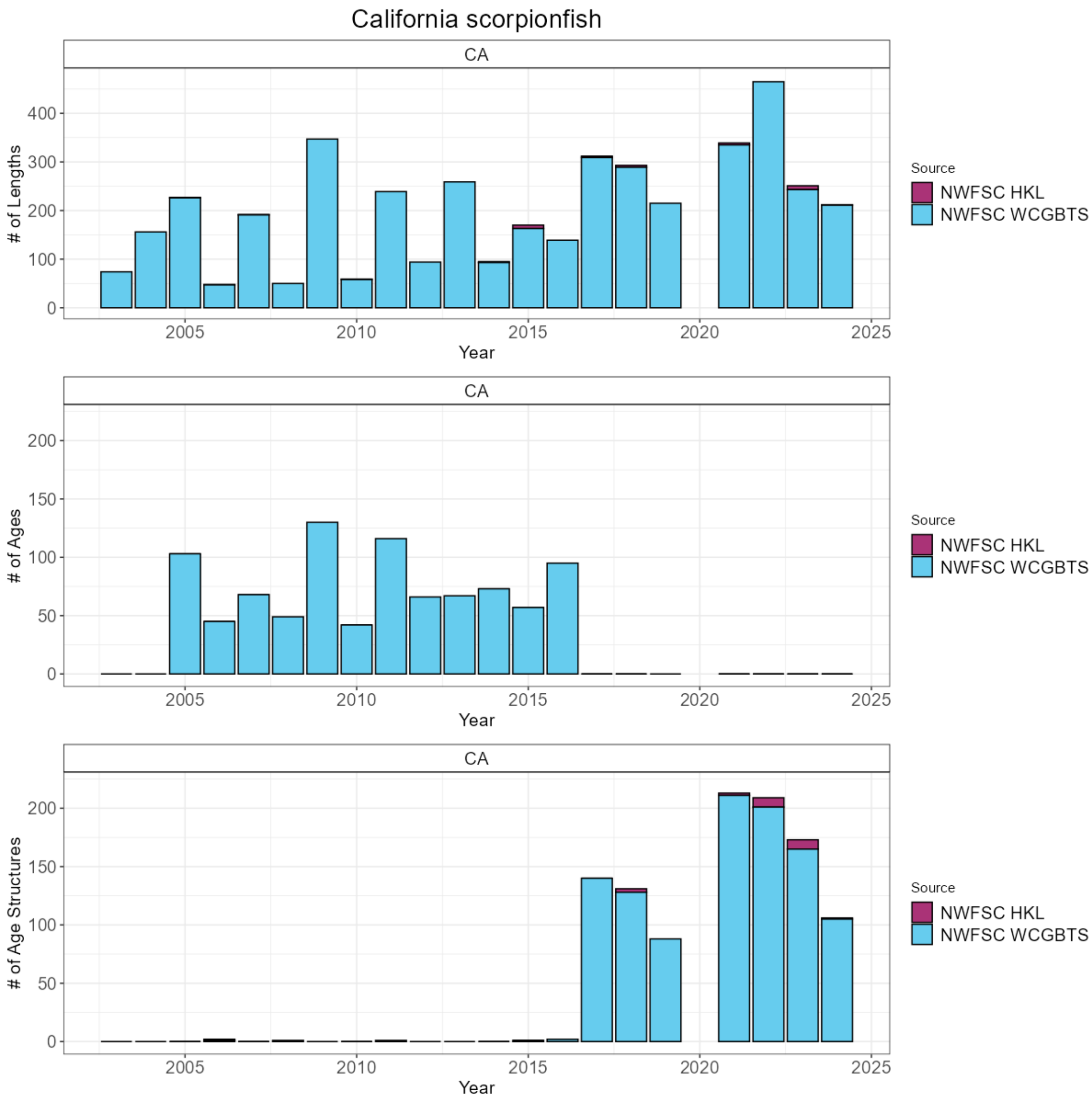

Figure 7: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.8 Canary rockfish

The most recent assessment of canary rockfish was a benchmark assessment conducted in 2023. Across available data, canary rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 7 positive tows per year. The NWFSC WCGBTS survey has an average of 51 positive tows per year. Across the coast a total of 1480 maturity samples have been collected with 1184 read maturities. ODFW scientists are planning a research cruise to Cobb Sea Mount in the summer of 2026 to sample age and sex distributions of rockfish species.

Table 8: Total number of available lengths, read ages, and unread age structures by data source and state between for canary rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	553	511	38
CA	NWFSC WCGBTS	3,397	1,854	190
OR	NWFSC WCGBTS	4,893	3,178	138
WA	NWFSC WCGBTS	6,382	3,430	377

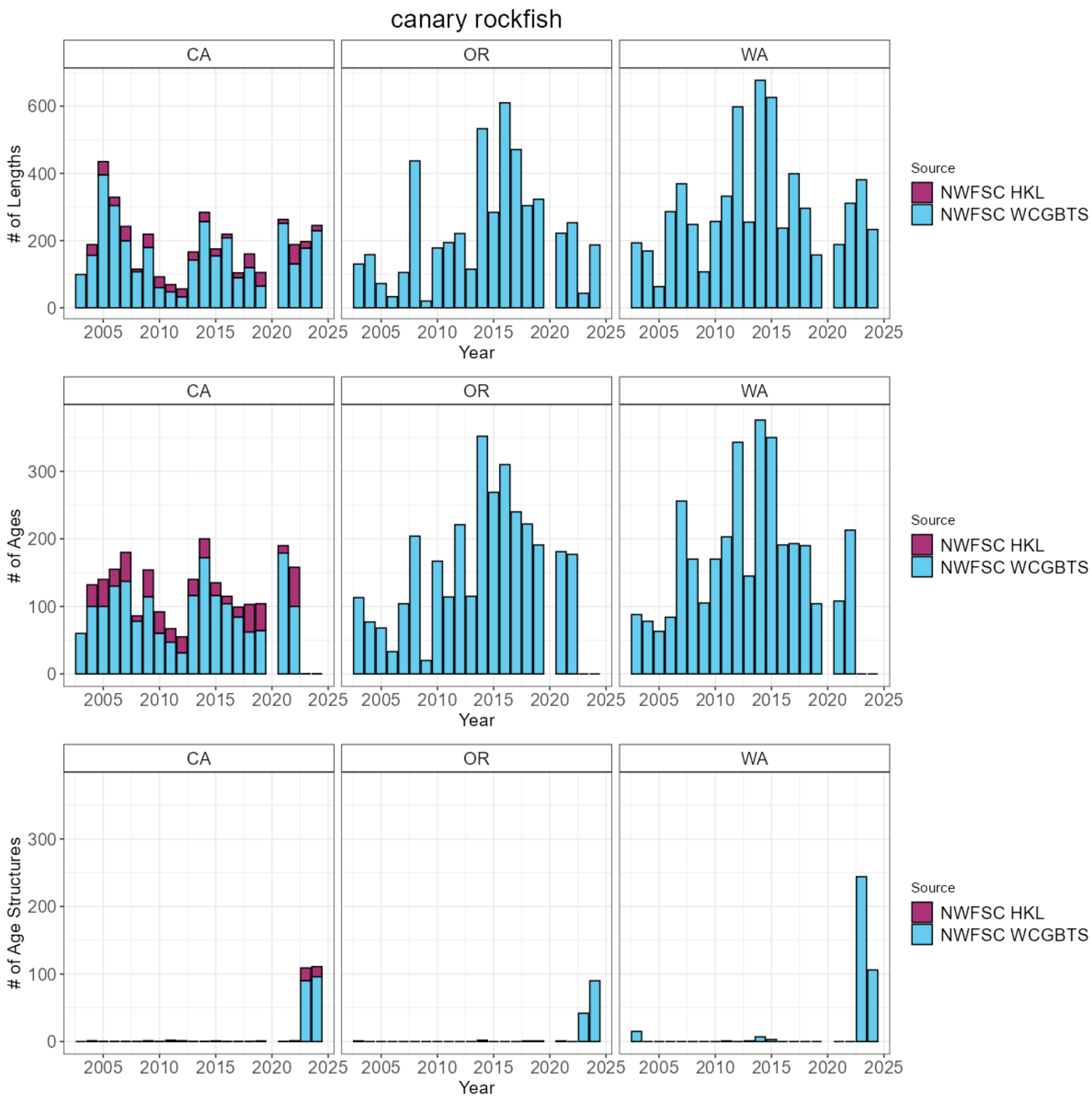

Figure 8: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.9 Chilipepper

The most recent assessment of chilipepper was a benchmark assessment conducted in 2025. Across available data, chilipepper have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 25 positive tows per year. The NWFSC WCGBTS survey has an average of 88 positive tows per year. Across the coast a total of 157 maturity samples have been collected with 157 read maturities.

Table 9: Total number of available lengths, read ages, and unread age structures by data source and state between for chilipepper.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	2,881	249	2,449
CA	NWFSC WCGBTS	33,744	12,386	728
OR	NWFSC WCGBTS	2,182	904	57
WA	NWFSC WCGBTS	27	27	0

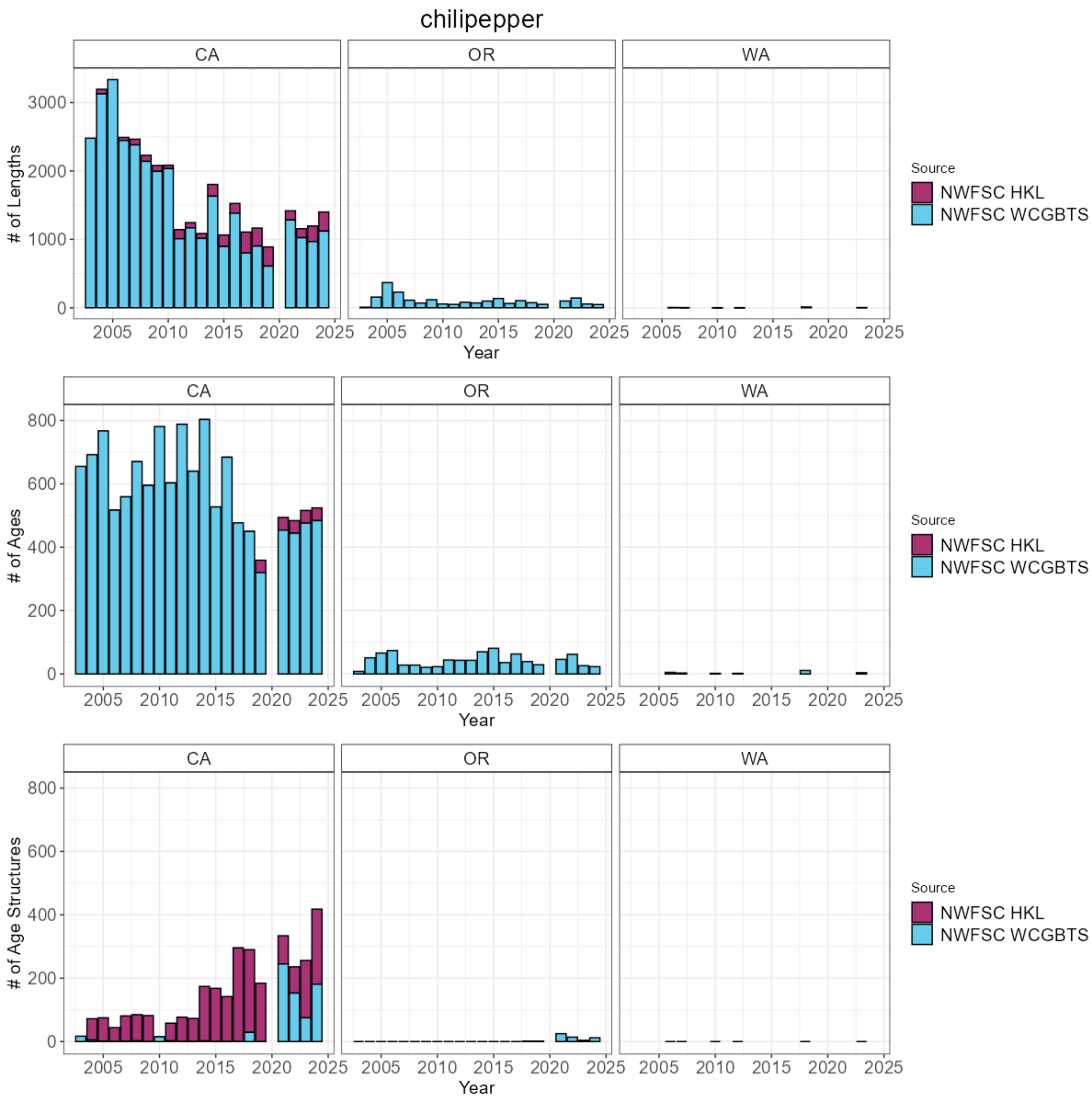

Figure 9: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.10 Darkblotched rockfish

The most recent assessment of darkblotched rockfish was an update assessment conducted in 2017. Across available data, darkblotched rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 107 positive tows per year. Across the coast a total of 958 maturity samples have been collected with 927 read maturities.

Table 10: Total number of available lengths, read ages, and unread age structures by data source and state between for darkblotched rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	7,634	2,561	925
OR	NWFSC WCGBTS	22,610	6,636	2,941
WA	NWFSC WCGBTS	8,474	2,523	789

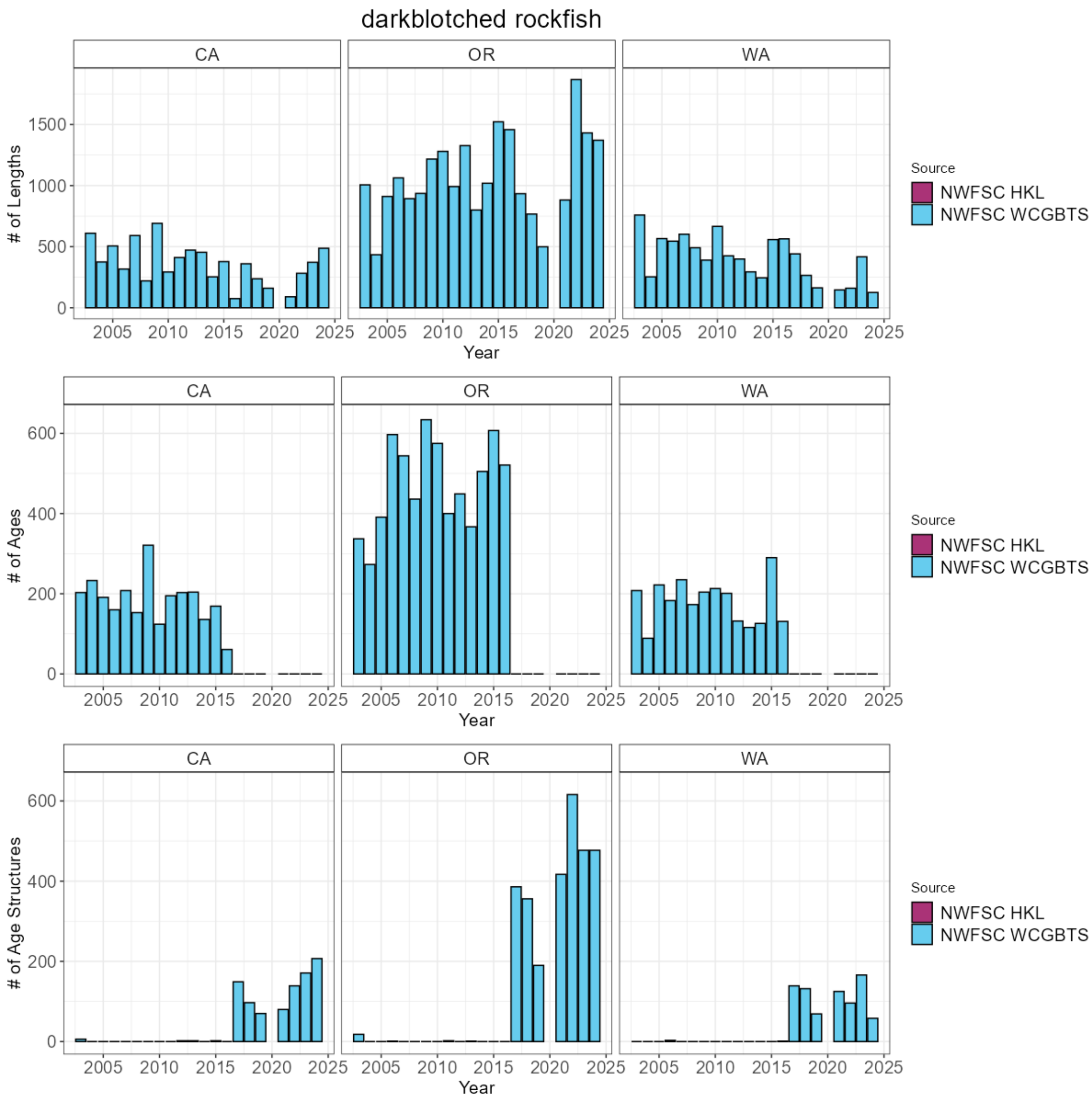

Figure 10: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.11 Dover sole

The most recent assessment of Dover sole was a benchmark assessment conducted in 2021. Across available data, Dover sole have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 508 positive tows per year. Across the coast a total of 820 maturity samples have been collected with 635 read maturities.

Table 11: Total number of available lengths, read ages, and unread age structures by data source and state between for Dover sole.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	86,127	8,700	5,775
OR	NWFSC WCGBTS	69,163	5,836	4,155
WA	NWFSC WCGBTS	32,188	2,581	1,982

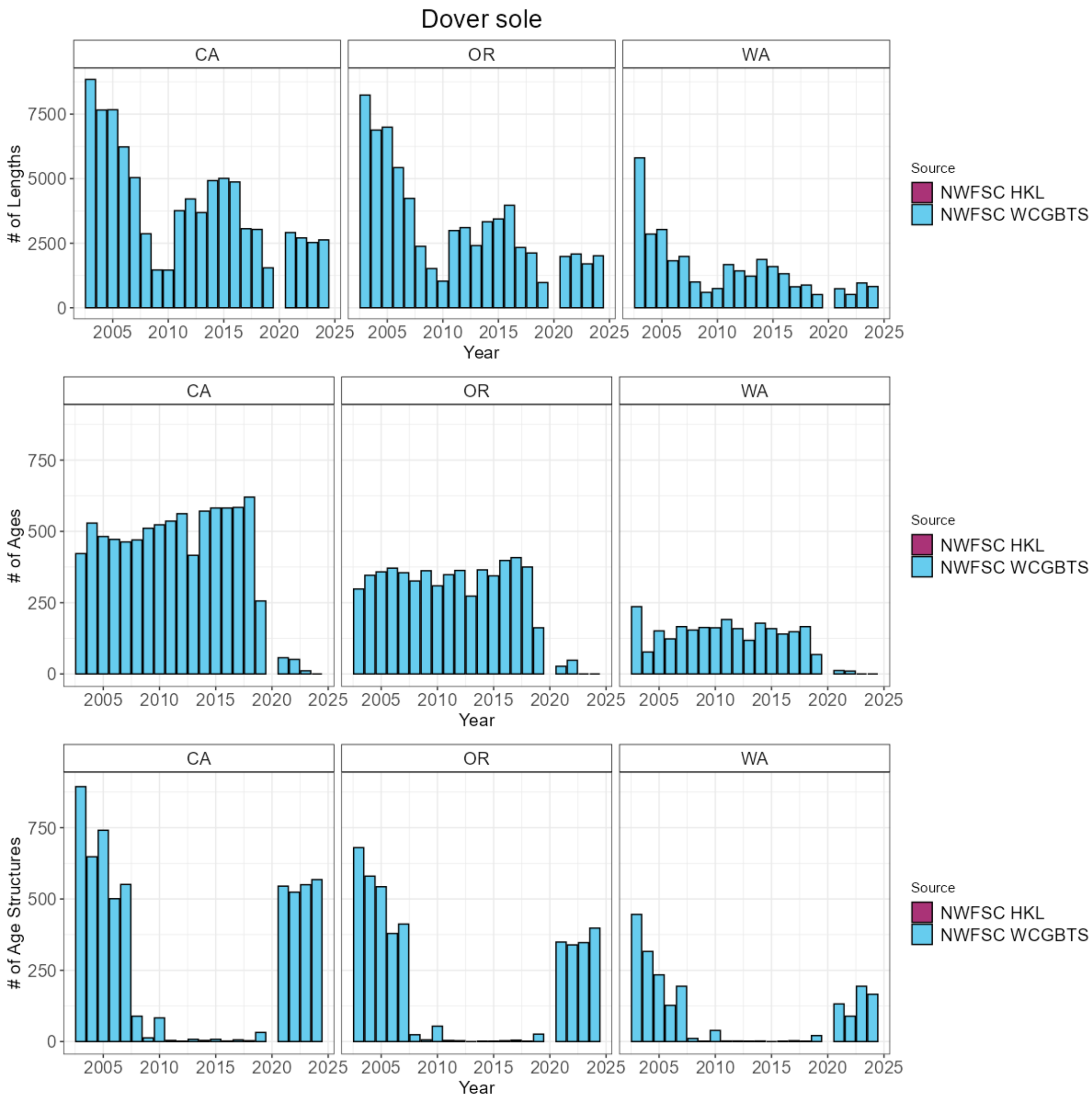

Figure 11: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.12 English sole

The most recent assessment of English sole was a data-moderate assessment conducted in 2013. Across available data, English sole have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 253 positive tows per year. Across the coast a total of 174 maturity samples have been collected with 0 read maturities.

Table 12: Total number of available lengths, read ages, and unread age structures by data source and state between for English sole.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	44,691	478	8,824
OR	NWFSC WCGBTS	31,096	279	5,347
WA	NWFSC WCGBTS	13,520	141	2,882

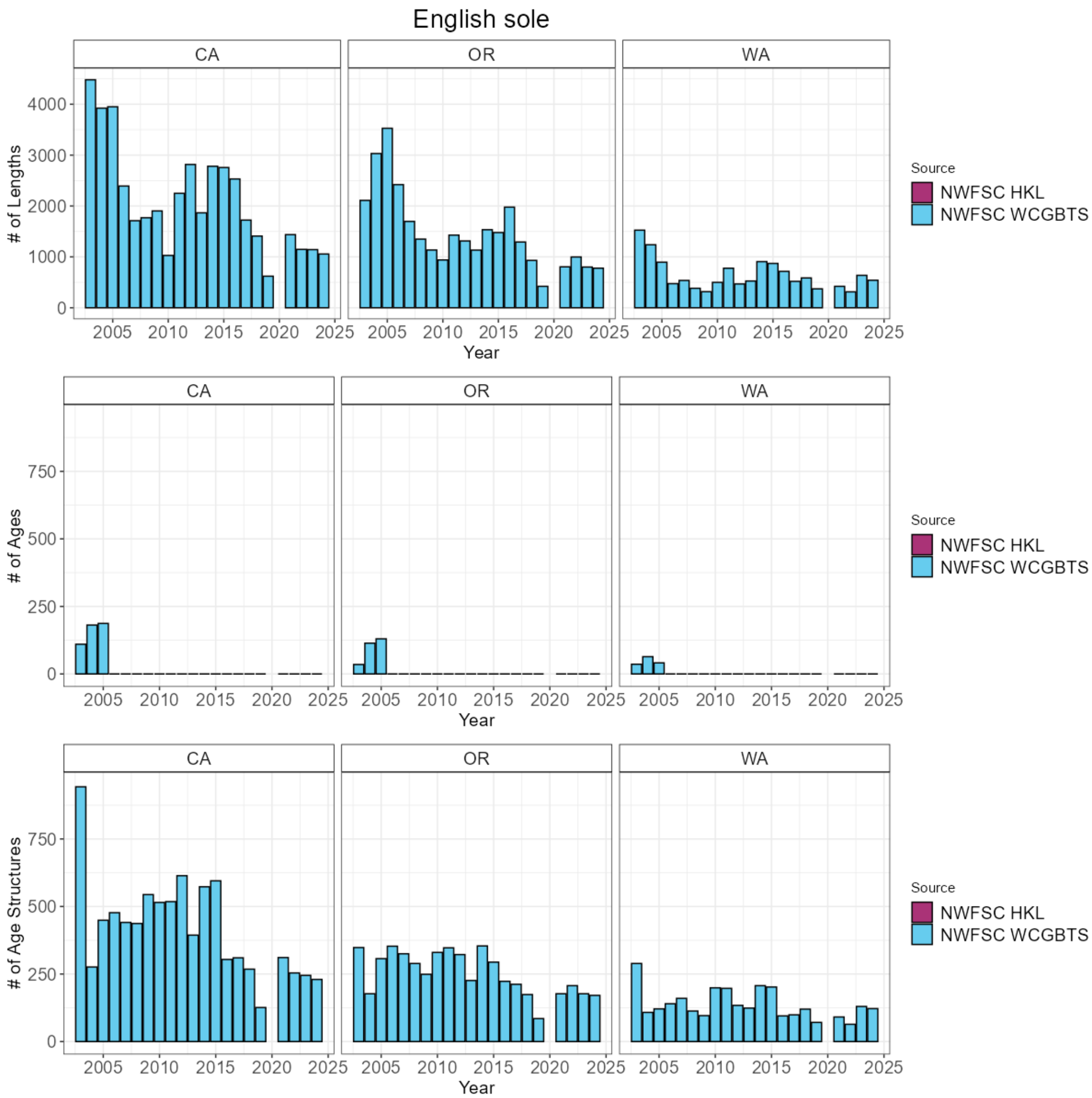

Figure 12: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.13 Flathead sole

To date, no assessment or analysis has been conducted on flathead sole. Across available data, flathead sole have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 49 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 13: Total number of available lengths, read ages, and unread age structures by data source and state between for flathead sole.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	76	0	38
OR	NWFSC WCGBTS	4,776	0	2,009
WA	NWFSC WCGBTS	7,168	0	1,639

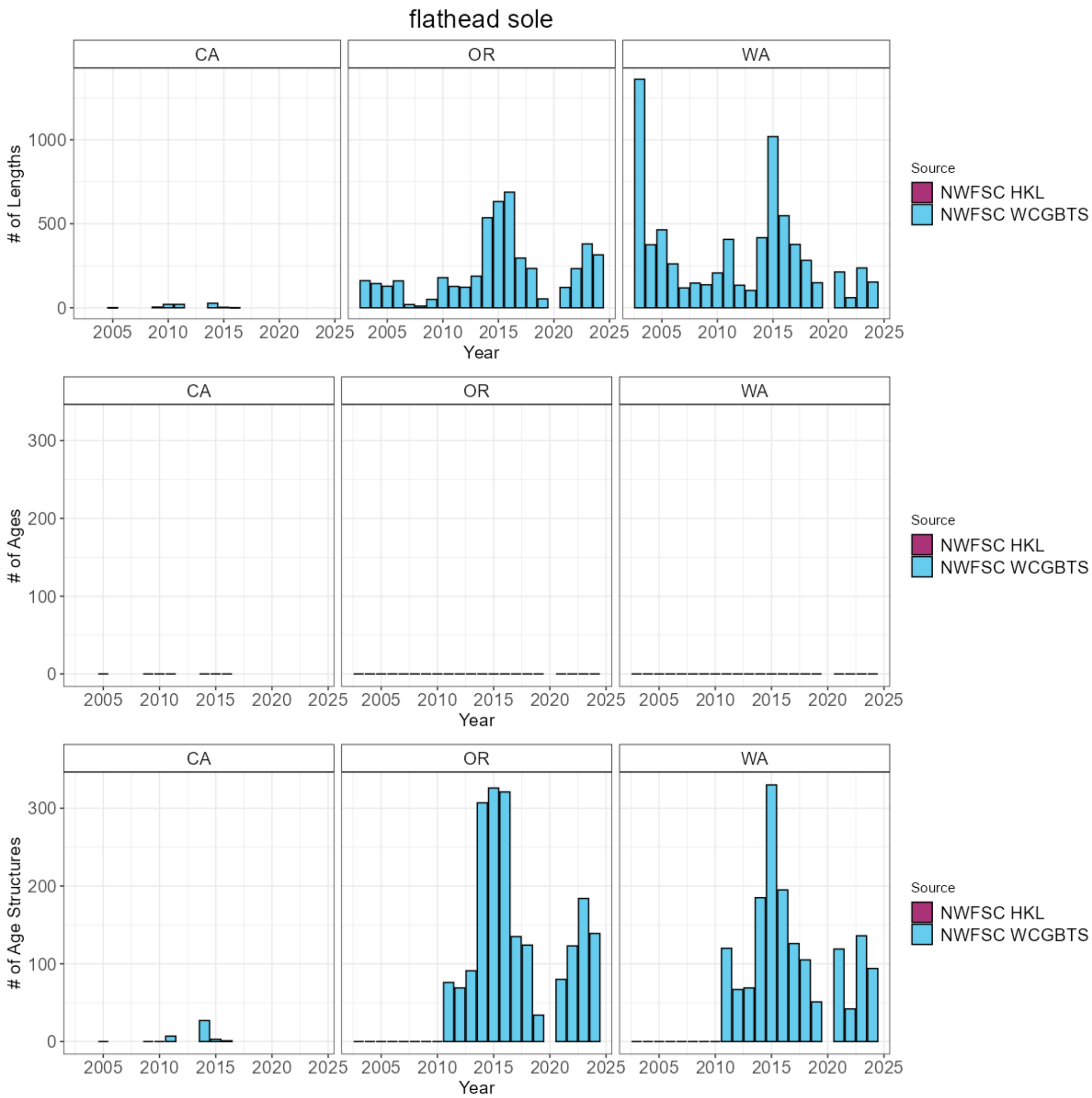

Figure 13: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.14 Greenspotted rockfish

The most recent assessment of greenspotted rockfish was a benchmark assessment conducted in 2011. Across available data, greenspotted rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 61 positive tows per year. The NWFSC WCGBTS survey has an average of 33 positive tows per year. Across the coast a total of 338 maturity samples have been collected with 175 read maturities.

Table 14: Total number of available lengths, read ages, and unread age structures by data source and state between for greenspotted rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	6,110	843	5,090
CA	NWFSC WCGBTS	7,797	701	3,823
OR	NWFSC WCGBTS	1,323	0	1,031
WA	NWFSC WCGBTS	40	0	37

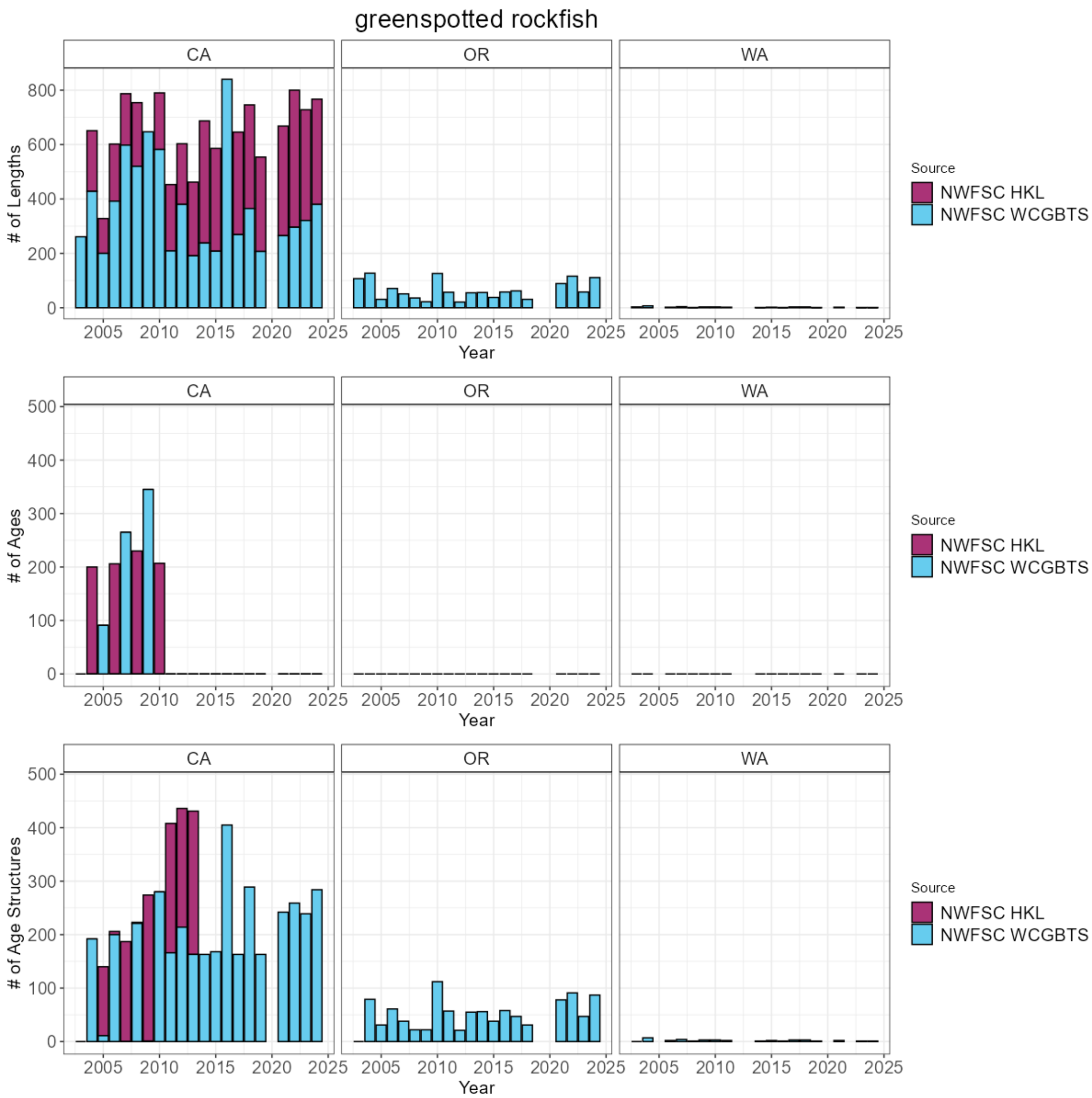

Figure 14: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.15 Greenstriped rockfish

The most recent assessment of greenstriped rockfish was a benchmark assessment conducted in 2009. Across available data, greenstriped rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 25 positive tows per year. The NWFSC WCGBTS survey has an average of 154 positive tows per year. Across the coast a total of 73 maturity samples have been collected with 73 read maturities.

Table 15: Total number of available lengths, read ages, and unread age structures by data source and state between for greenstriped rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	1,180	0	1,167
CA	NWFSC WCGBTS	16,867	1,359	4,220
OR	NWFSC WCGBTS	19,607	1,346	3,924
WA	NWFSC WCGBTS	10,355	709	1,918

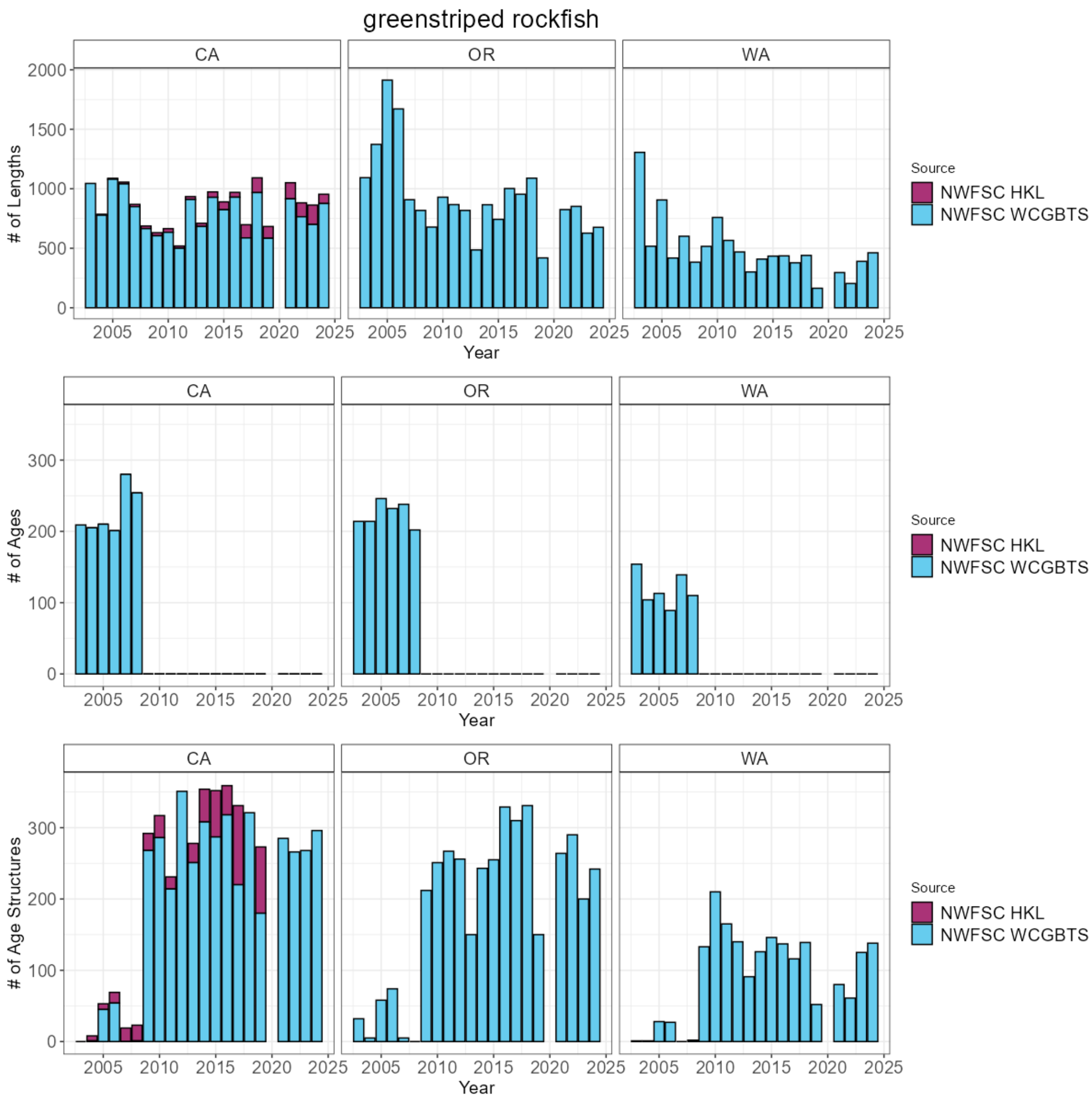

Figure 15: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.16 Kelp greenling

The most recent assessment of kelp greenling was a benchmark assessment conducted in 2015. Across available data, kelp greenling have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 9 positive tows per year. Across the coast a total of 8 maturity samples have been collected with 8 read maturities. ODFW scientists have conducted research looking at the influences of hypoxia on the catch per unit effort for kelp greenling.

Table 16: Total number of available lengths, read ages, and unread age structures by data source and state between for kelp greenling.

State	Source	Lengths	Ages	Age Structures
OR	NWFSC WCGBTS	673	0	504
WA	NWFSC WCGBTS	214	0	194

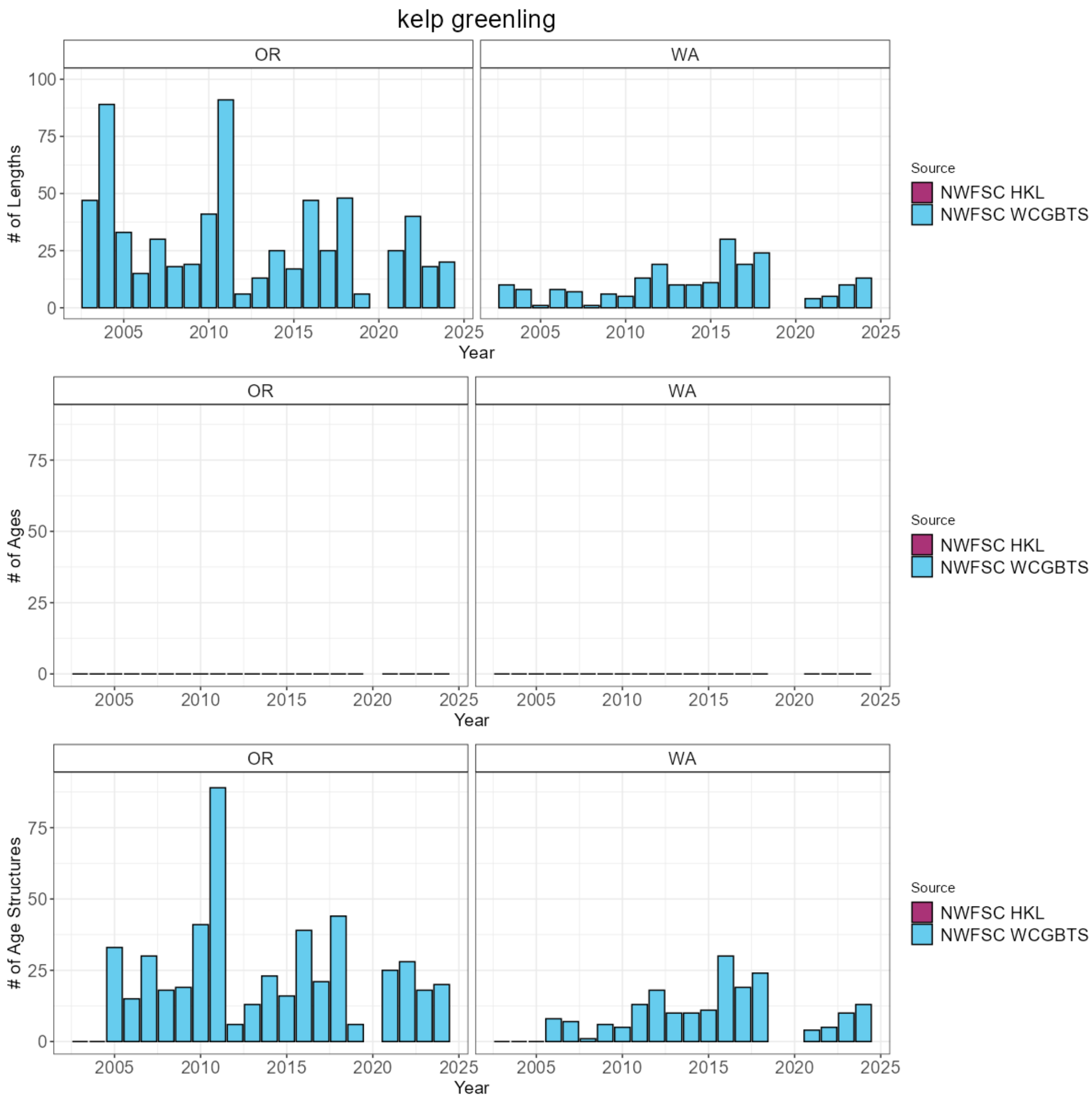

Figure 16: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.17 Lingcod north

The most recent assessment of lingcod north was a benchmark assessment conducted in 2021. Across available data, lingcod north have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 124 positive tows per year. Across the coast a total of 1161 maturity samples have been collected with 1031 read maturities. ODFW scientists have conducted research looking at the influences of hypoxia on the catch per unit effort for lingcod north.

Table 17: Total number of available lengths, read ages, and unread age structures by data source and state between for lingcod north.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	2,523	780	597
OR	NWFSC WCGBTS	9,519	2,782	2,524
WA	NWFSC WCGBTS	6,684	1,582	1,345

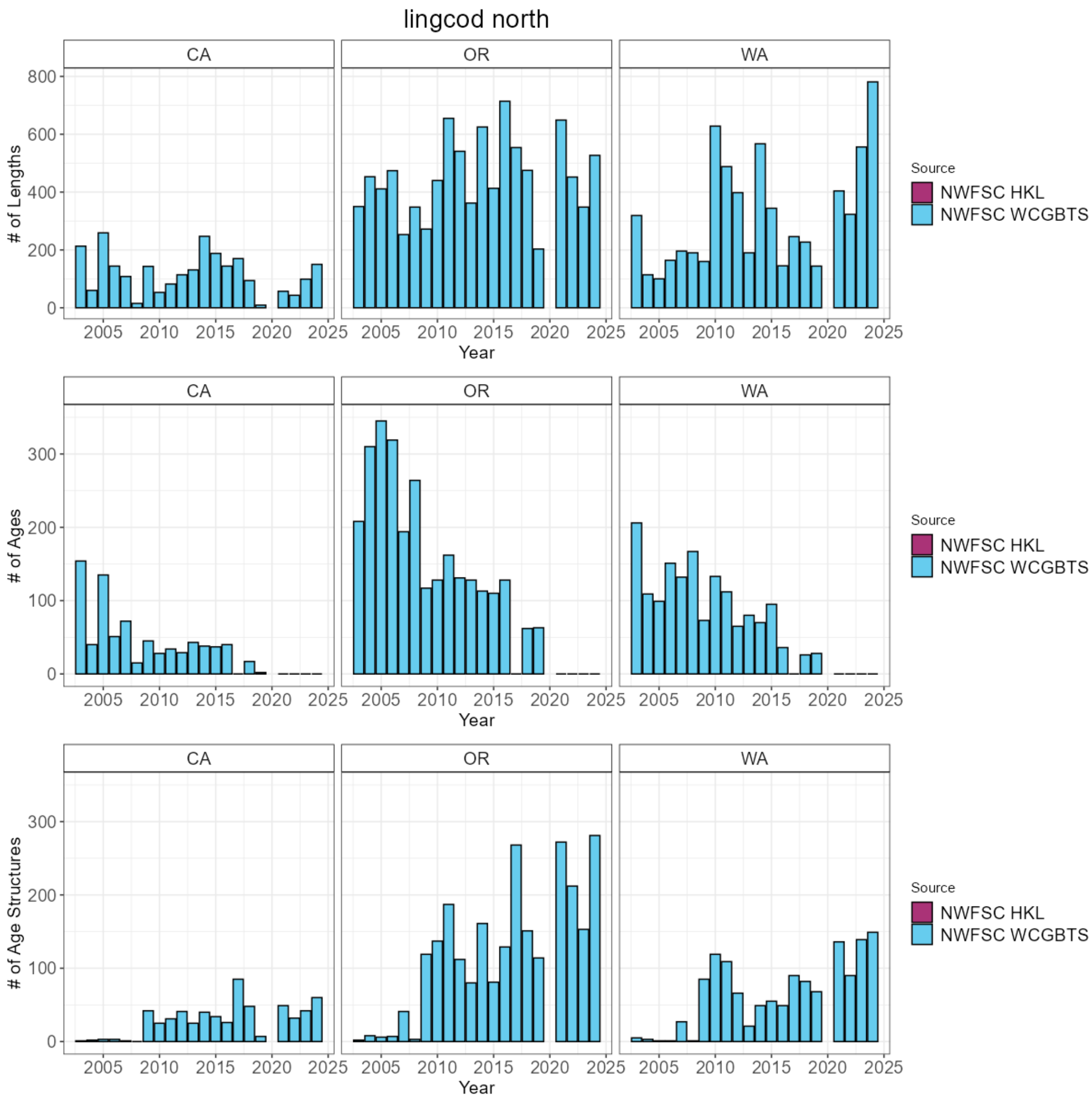

Figure 17: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.18 Lingcod south

The most recent assessment of lingcod south was a benchmark assessment conducted in 2021. Across available data, lingcod south have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 75 positive tows per year. Across the coast a total of 1161 maturity samples have been collected with 1031 read maturities.

Table 18: Total number of available lengths, read ages, and unread age structures by data source and state between for lingcod south.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	11,921	3,870	2,491

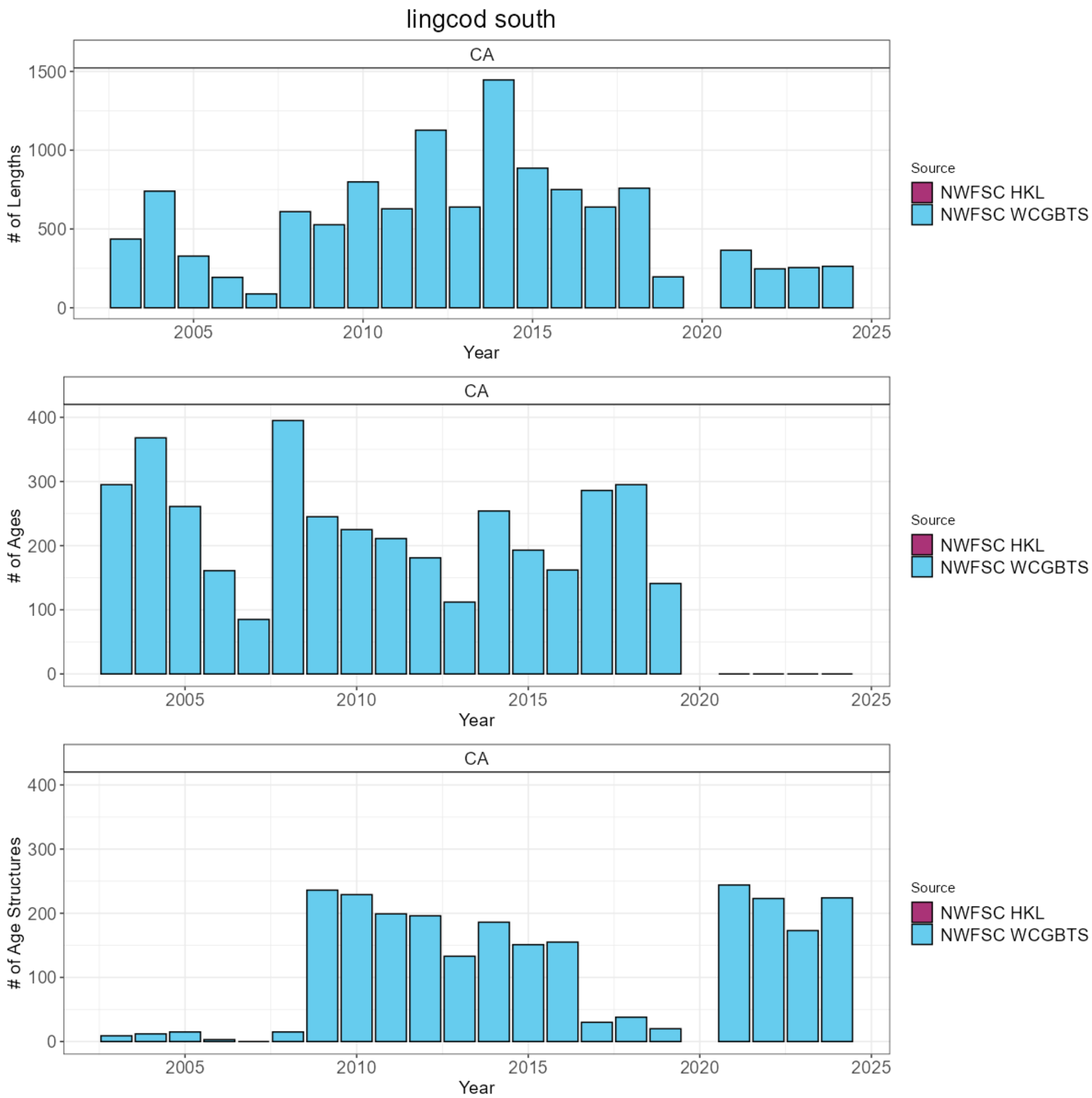

Figure 18: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.19 Longnose skate

The most recent assessment of longnose skate was a benchmark assessment conducted in 2019. Across available data, longnose skate have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 364 positive tows per year. Across the coast a total of 508 maturity samples have been collected with 508 read maturities.

Table 19: Total number of available lengths, read ages, and unread age structures by data source and state between for longnose skate.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	34,520	336	1,229
OR	NWFSC WCGBTS	18,099	227	850
WA	NWFSC WCGBTS	8,858	84	347

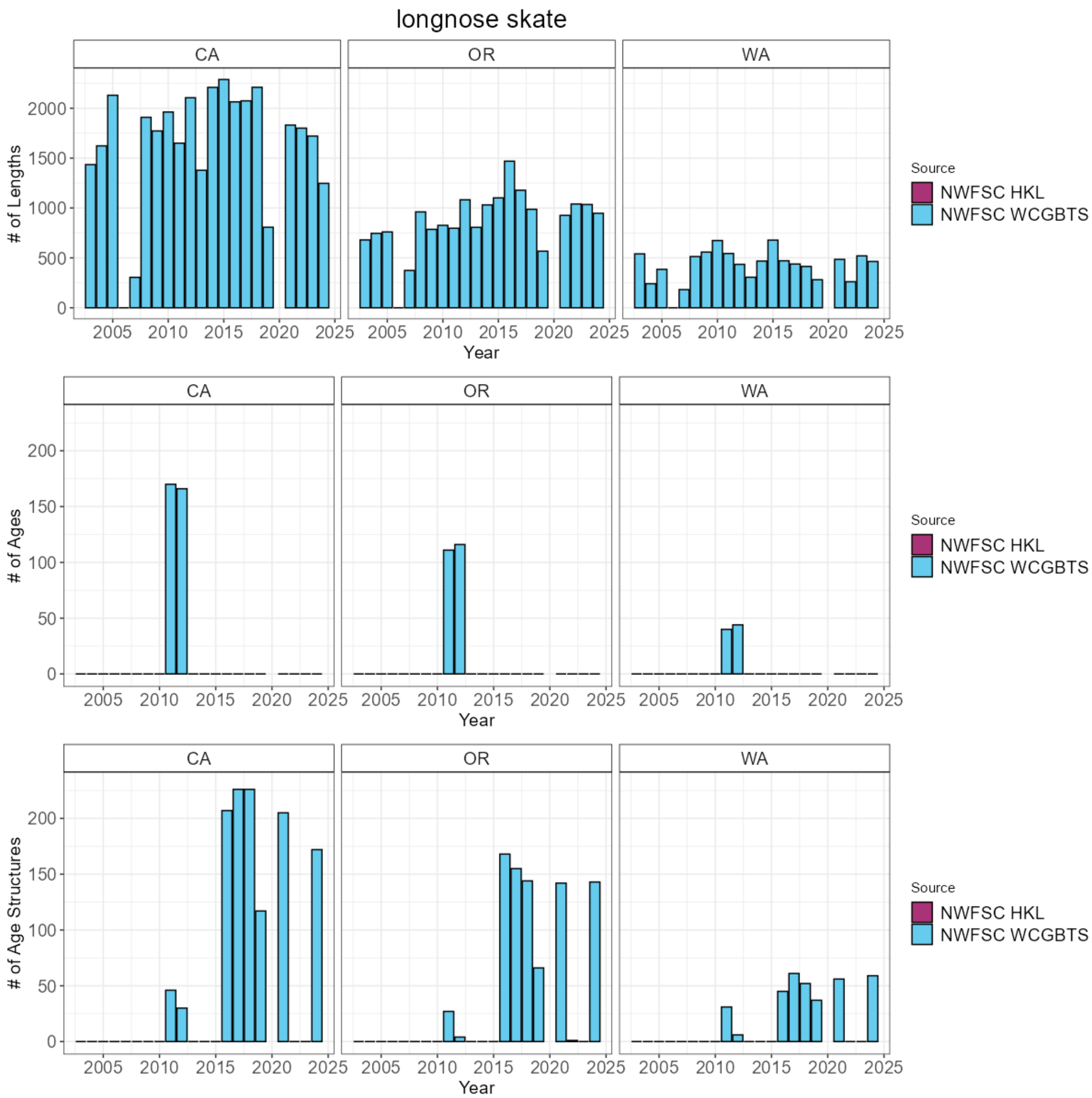

Figure 19: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.20 Longspine thornyhead

The most recent assessment of longspine thornyhead was a benchmark assessment conducted in 2013. Across available data, longspine thornyhead have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 211 positive tows per year. Across the coast a total of 341 maturity samples have been collected with 63 read maturities.

Table 20: Total number of available lengths, read ages, and unread age structures by data source and state between for longspine thornyhead.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	73,218	0	9,607
OR	NWFSC WCGBTS	33,808	0	4,111
WA	NWFSC WCGBTS	14,270	0	1,777

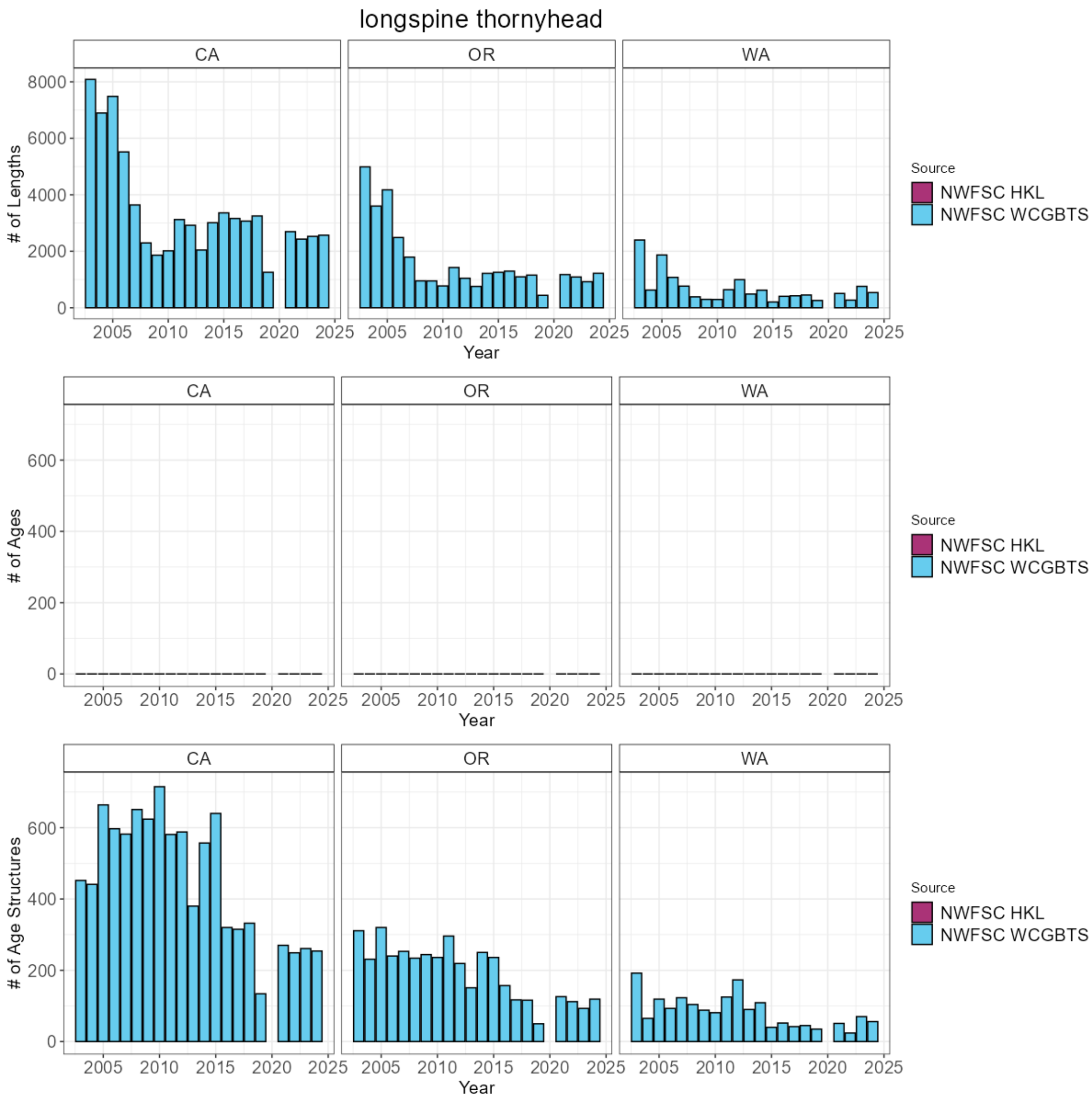

Figure 20: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.21 Pacific cod

To date, no assessment or analysis has been conducted on Pacific cod. Across available data, Pacific cod have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 27 positive tows per year. Across the coast a total of 125 maturity samples have been collected with 0 read maturities.

Table 21: Total number of available lengths, read ages, and unread age structures by data source and state between for Pacific cod.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	14	0	0
OR	NWFSC WCGBTS	511	0	215
WA	NWFSC WCGBTS	3,747	0	1,342

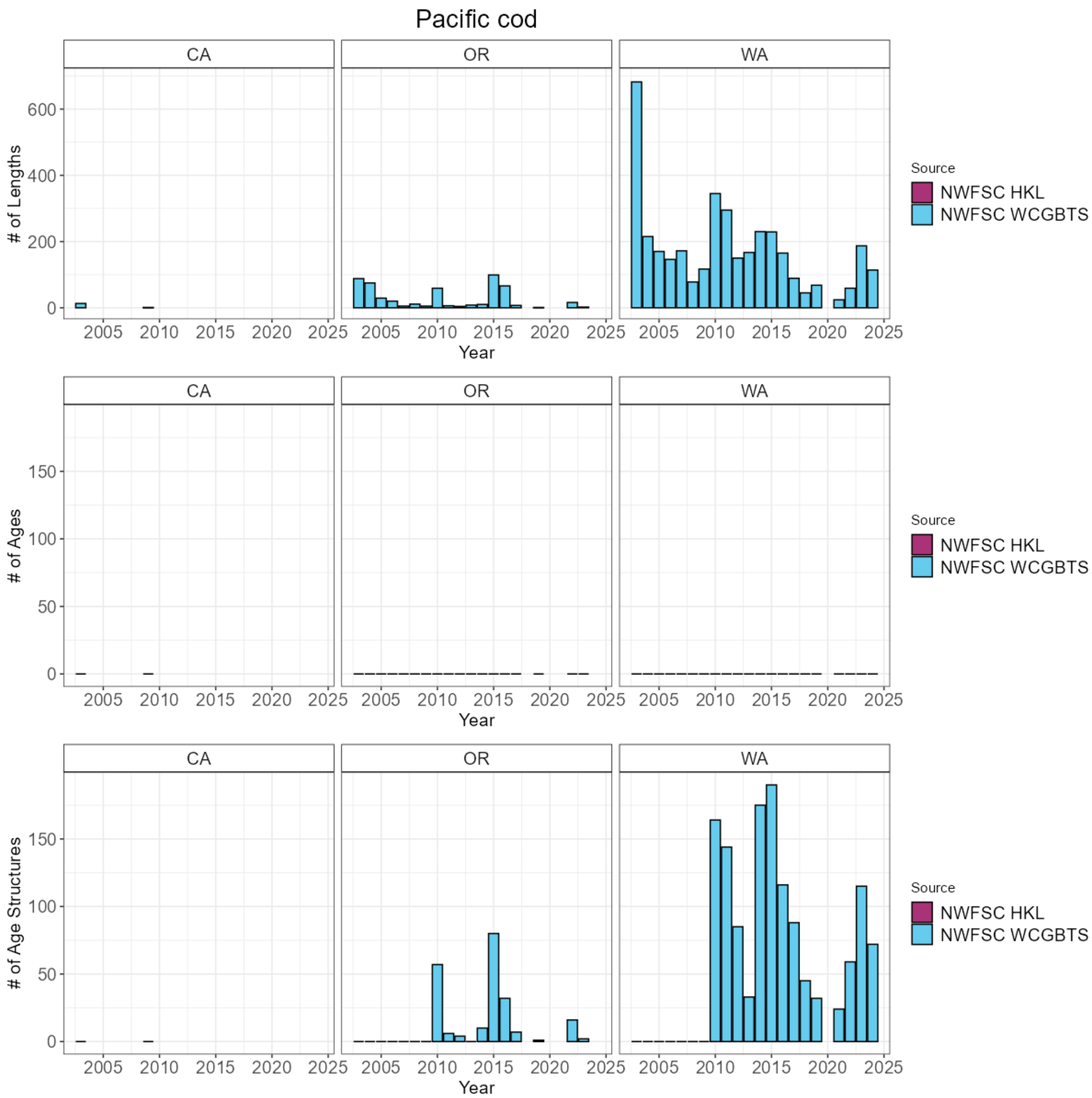

Figure 21: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.22 Pacific ocean perch

The most recent assessment of Pacific ocean perch was a benchmark assessment conducted in 2017. Across available data, Pacific ocean perch have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 46 positive tows per year. Across the coast a total of 583 maturity samples have been collected with 583 read maturities.

Table 22: Total number of available lengths, read ages, and unread age structures by data source and state between for Pacific ocean perch.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	232	78	153
OR	NWFSC WCGBTS	9,520	3,061	3,776
WA	NWFSC WCGBTS	8,009	2,743	1,955

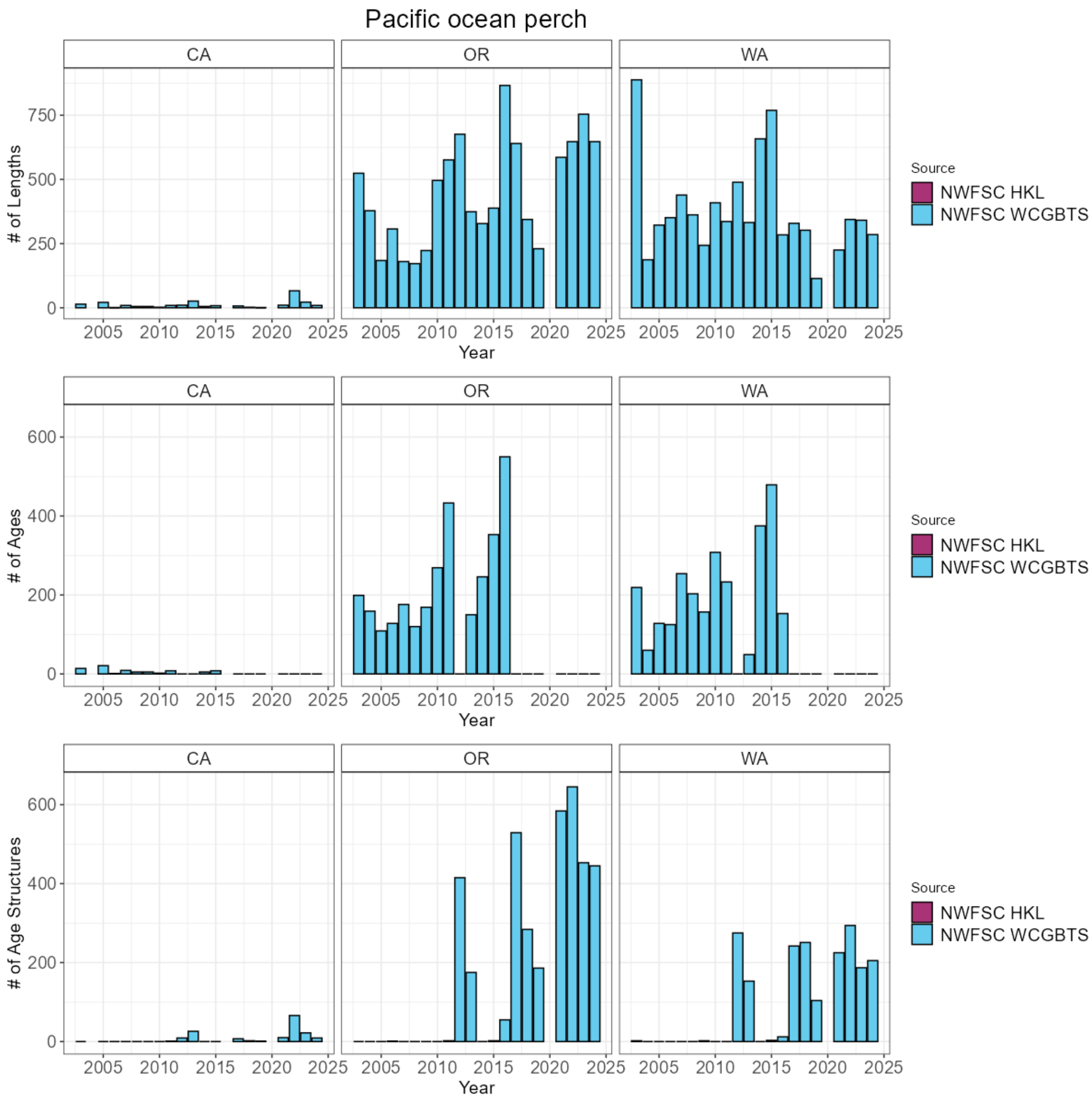

Figure 22: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.23 Pacific sanddab

To date, no assessment or analysis has been conducted on Pacific sanddab. Across available data, Pacific sanddab have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 198 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 23: Total number of available lengths, read ages, and unread age structures by data source and state between for Pacific sanddab.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	53,720	4,714	3,943
OR	NWFSC WCGBTS	27,649	2,374	2,175
WA	NWFSC WCGBTS	10,581	874	903

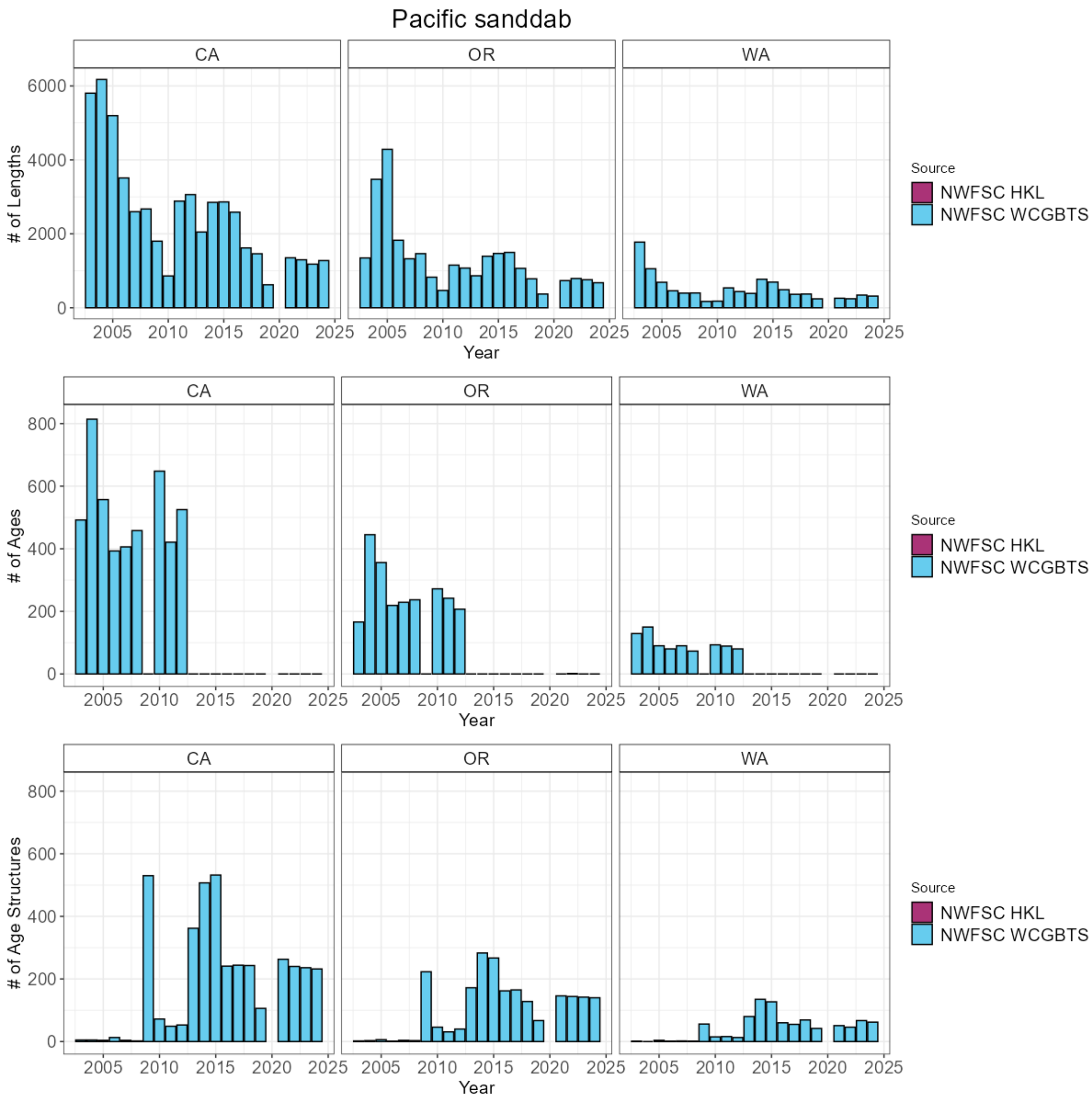

Figure 23: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.24 Pacific spiny dogfish

The most recent assessment of Pacific spiny dogfish was a benchmark assessment conducted in 2021. Across available data, Pacific spiny dogfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 155 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities. Researchers have conducted an analysis of the trends of Pacific spiny dogfish across the North Pacific indicating declining population trends across regions.

Table 24: Total number of available lengths, read ages, and unread age structures by data source and state between for Pacific spiny dogfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	7	0	0
CA	NWFSC WCGBTS	17,911	285	4,744
OR	NWFSC WCGBTS	4,384	128	1,879
WA	NWFSC WCGBTS	11,562	178	2,877

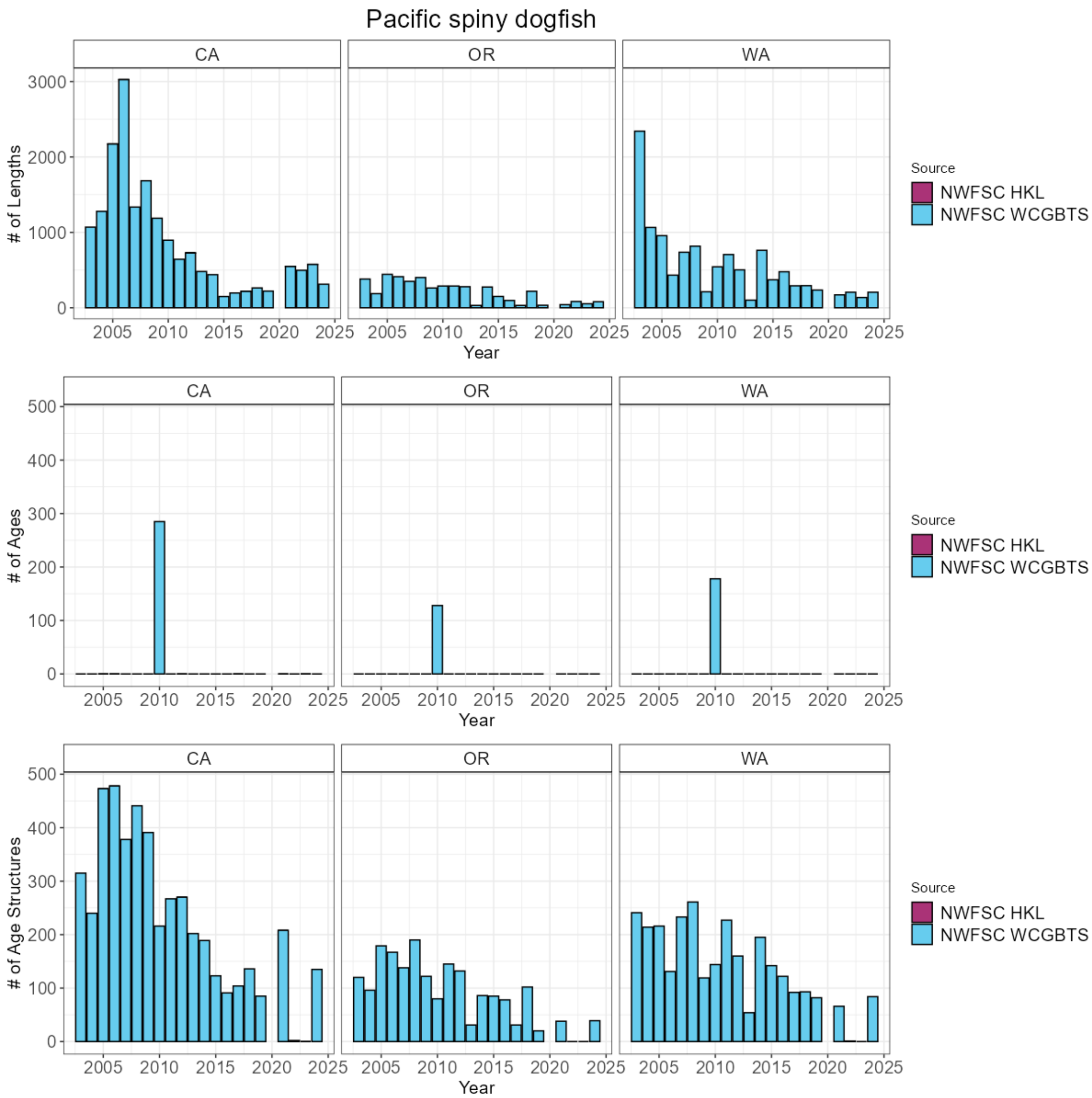

Figure 24: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.25 Petrale sole

The most recent assessment of petrale sole was a benchmark assessment conducted in 2023. Across available data, petrale sole have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 260 positive tows per year. Across the coast a total of 728 maturity samples have been collected with 598 read maturities.

Table 25: Total number of available lengths, read ages, and unread age structures by data source and state between for petrale sole.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	4	0	0
CA	NWFSC WCGBTS	35,677	7,165	3,016
OR	NWFSC WCGBTS	31,575	4,857	3,589
WA	NWFSC WCGBTS	15,349	2,676	2,089

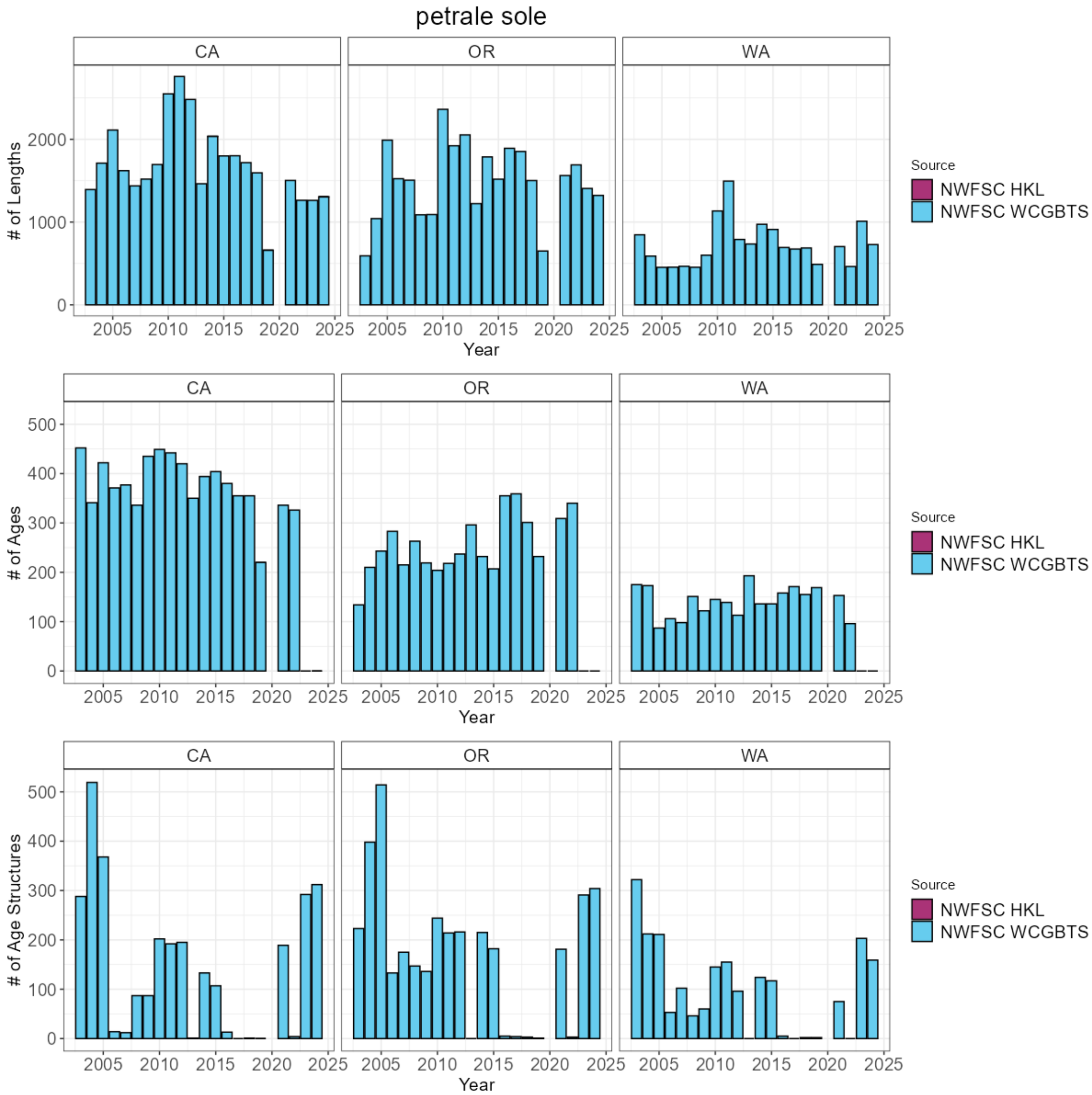

Figure 25: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.26 Quillback rockfish

The most recent assessment of quillback rockfish was a data-moderate assessment conducted in 2021. Across available data, quillback rockfish have been observed and sampled by the NWFSC WCGBT survey. Across the coast a total of 167 maturity samples have been collected with 162 read maturities. ODFW scientists have conducted research looking at the influences of hypoxia on the catch per unit effort for quillback rockfish.

Table 26: Total number of available lengths, read ages, and unread age structures by data source and state between for quillback rockfish.

State	Source	Lengths	Ages	Age Structures
OR	NWFSC WCGBTS	124	87	20
WA	NWFSC WCGBTS	105	75	9

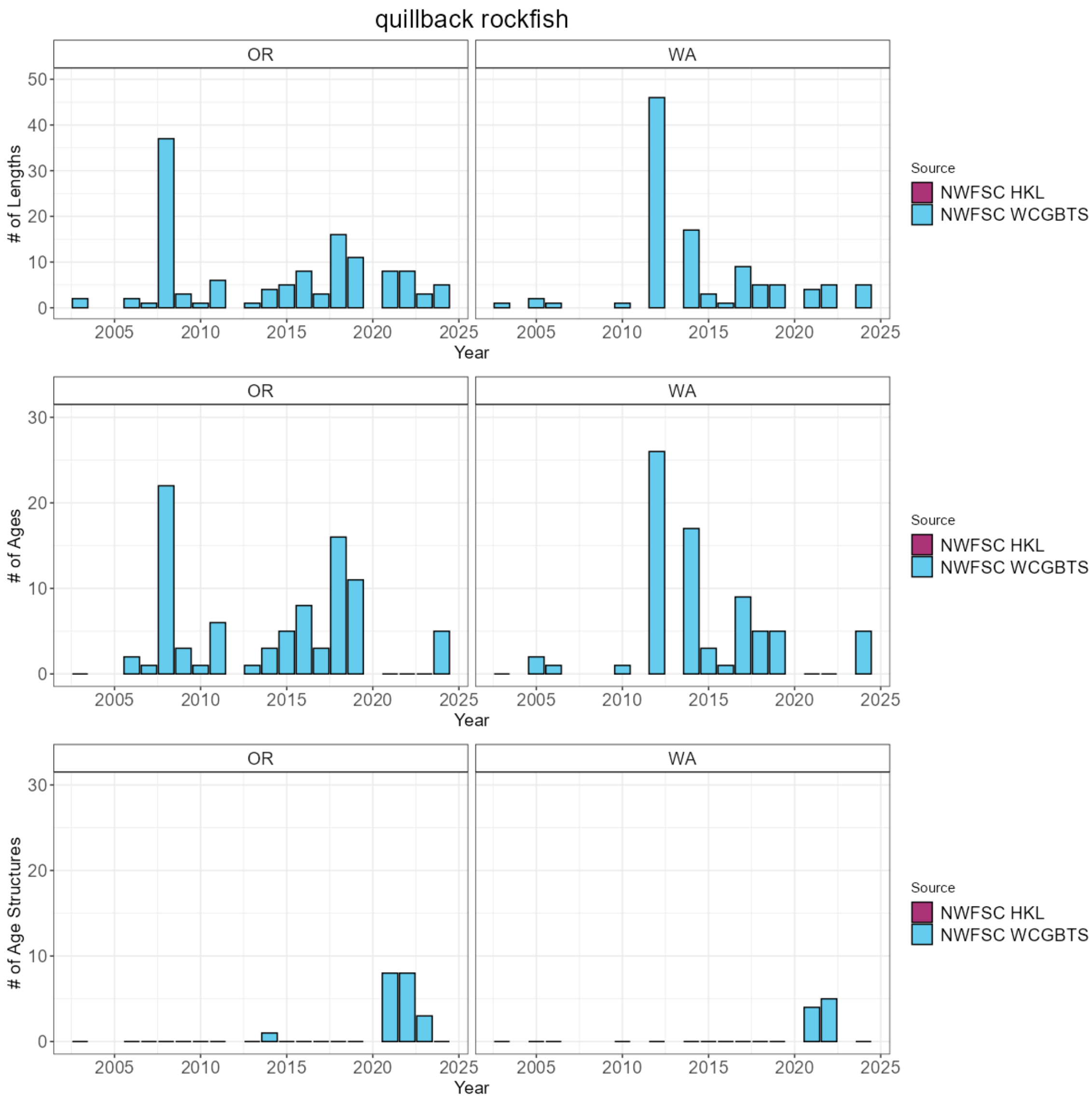

Figure 26: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.27 Redbanded rockfish

To date, no assessment or analysis has been conducted on redbanded rockfish. Across available data, redbanded rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 51 positive tows per year. Across the coast a total of 330 maturity samples have been collected with 125 read maturities.

Table 27: Total number of available lengths, read ages, and unread age structures by data source and state between for redbanded rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	1,014	0	982
OR	NWFSC WCGBTS	1,937	0	1,846
WA	NWFSC WCGBTS	1,091	0	1,018

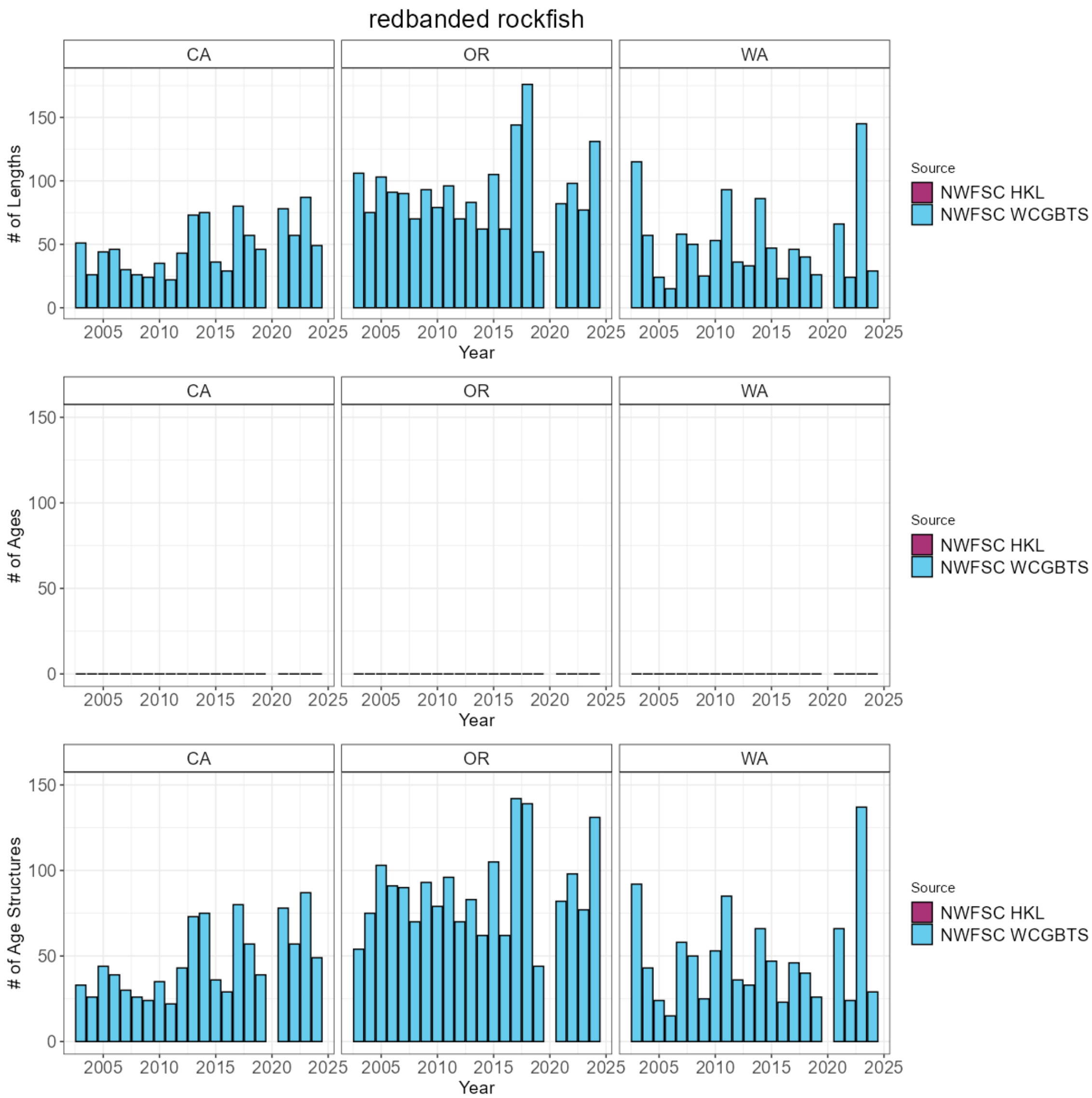

Figure 27: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.28 Redstripe rockfish

To date, no assessment or analysis has been conducted on redstripe rockfish. Across available data, redstripe rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 12 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 28: Total number of available lengths, read ages, and unread age structures by data source and state between for redstripe rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	348	0	199
OR	NWFSC WCGBTS	4,195	0	1,892
WA	NWFSC WCGBTS	3,276	0	1,646

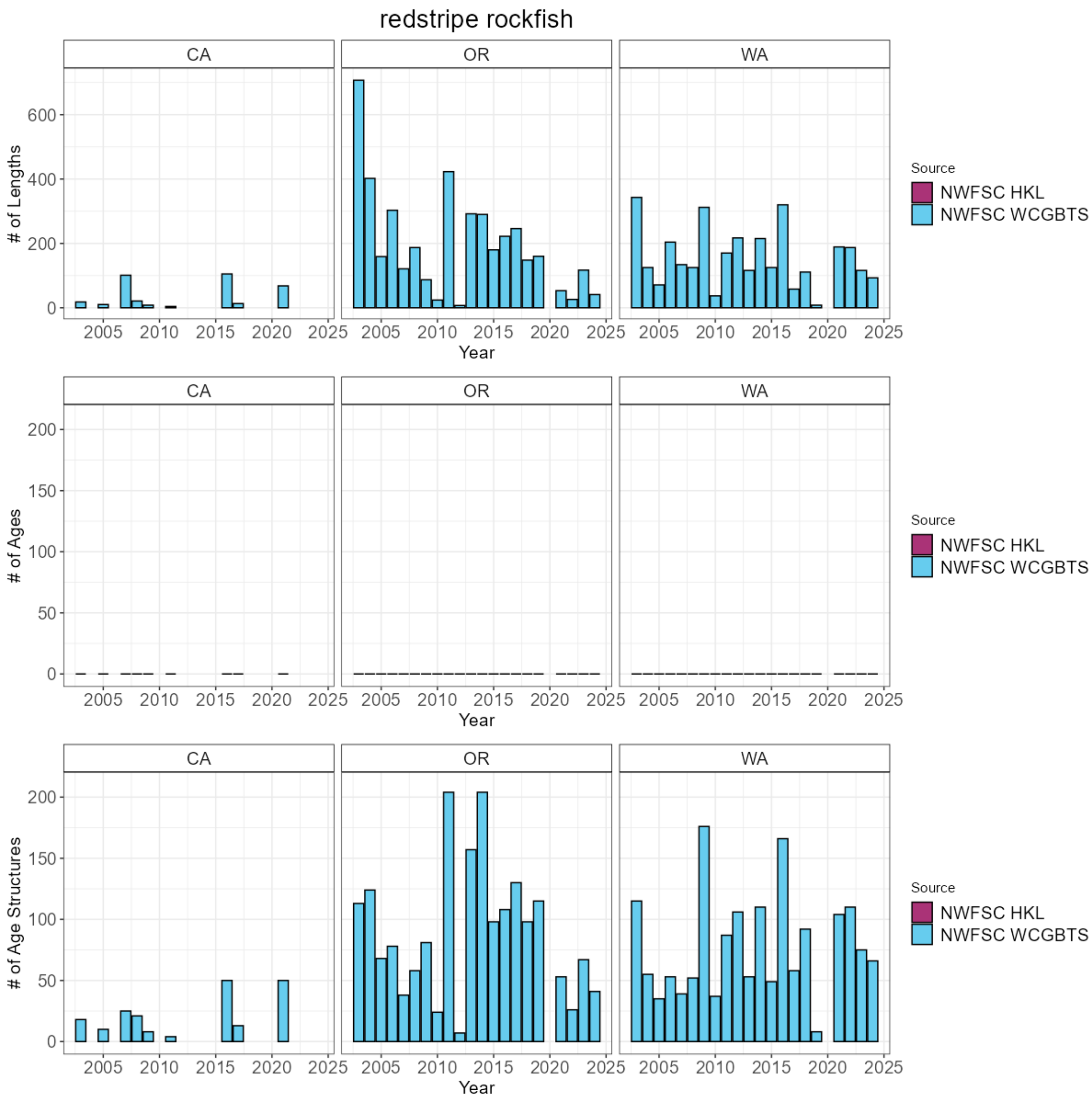

Figure 28: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.29 Rex sole

The most recent assessment of rex sole was a data-moderate assessment conducted in 2023. Across available data, rex sole have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 367 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 29: Total number of available lengths, read ages, and unread age structures by data source and state between for rex sole.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	57,526	273	4,920
OR	NWFSC WCGBTS	67,944	208	4,700
WA	NWFSC WCGBTS	25,156	140	1,917

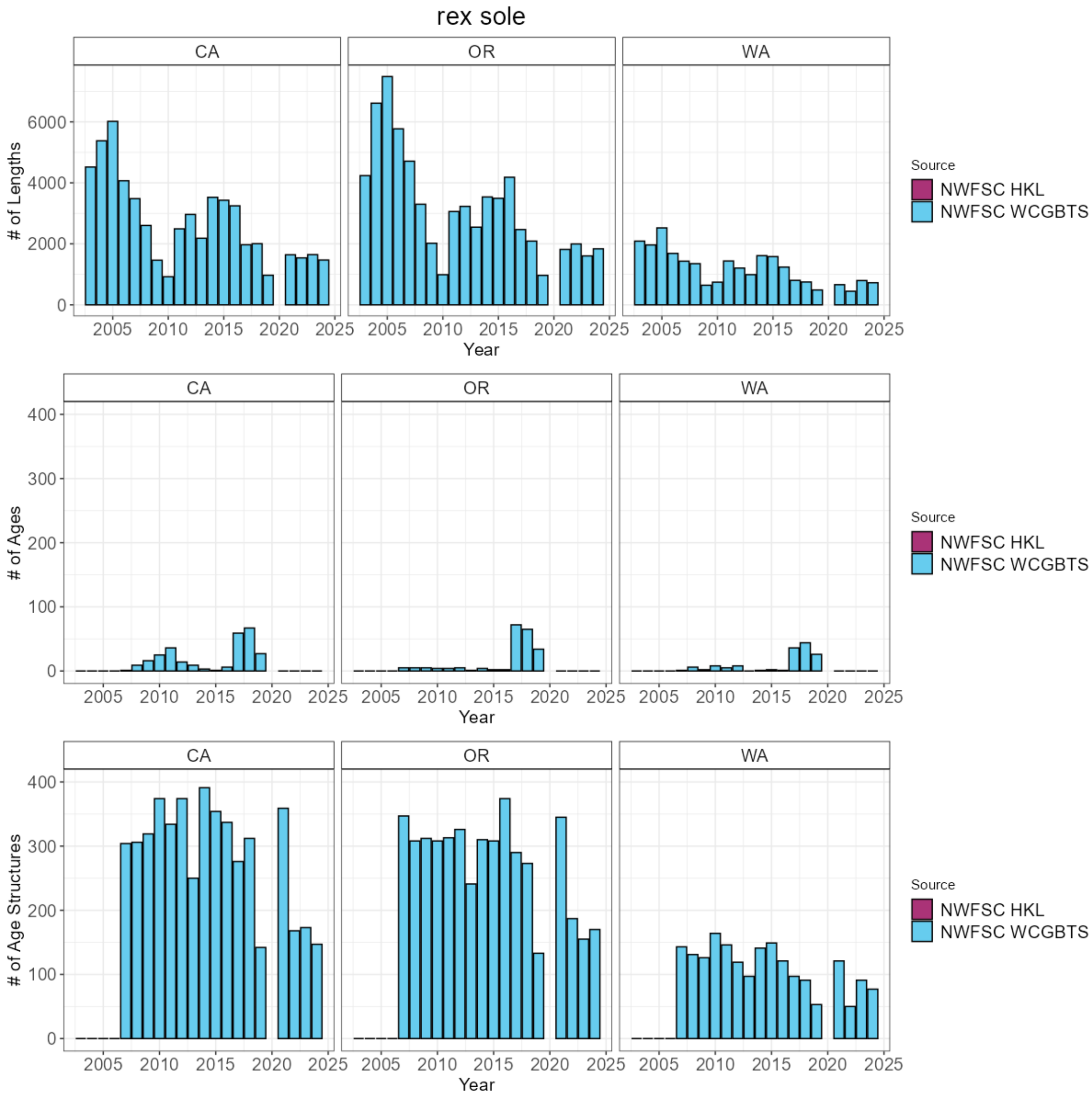

Figure 29: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.30 Rosethorn rockfish

To date, no assessment or analysis has been conducted on rosethorn rockfish. Across available data, rosethorn rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 46 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 30: Total number of available lengths, read ages, and unread age structures by data source and state between for rosethorn rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	36	0	27
CA	NWFSC WCGBTS	3,060	0	1,528
OR	NWFSC WCGBTS	10,290	0	4,214
WA	NWFSC WCGBTS	6,822	0	2,827

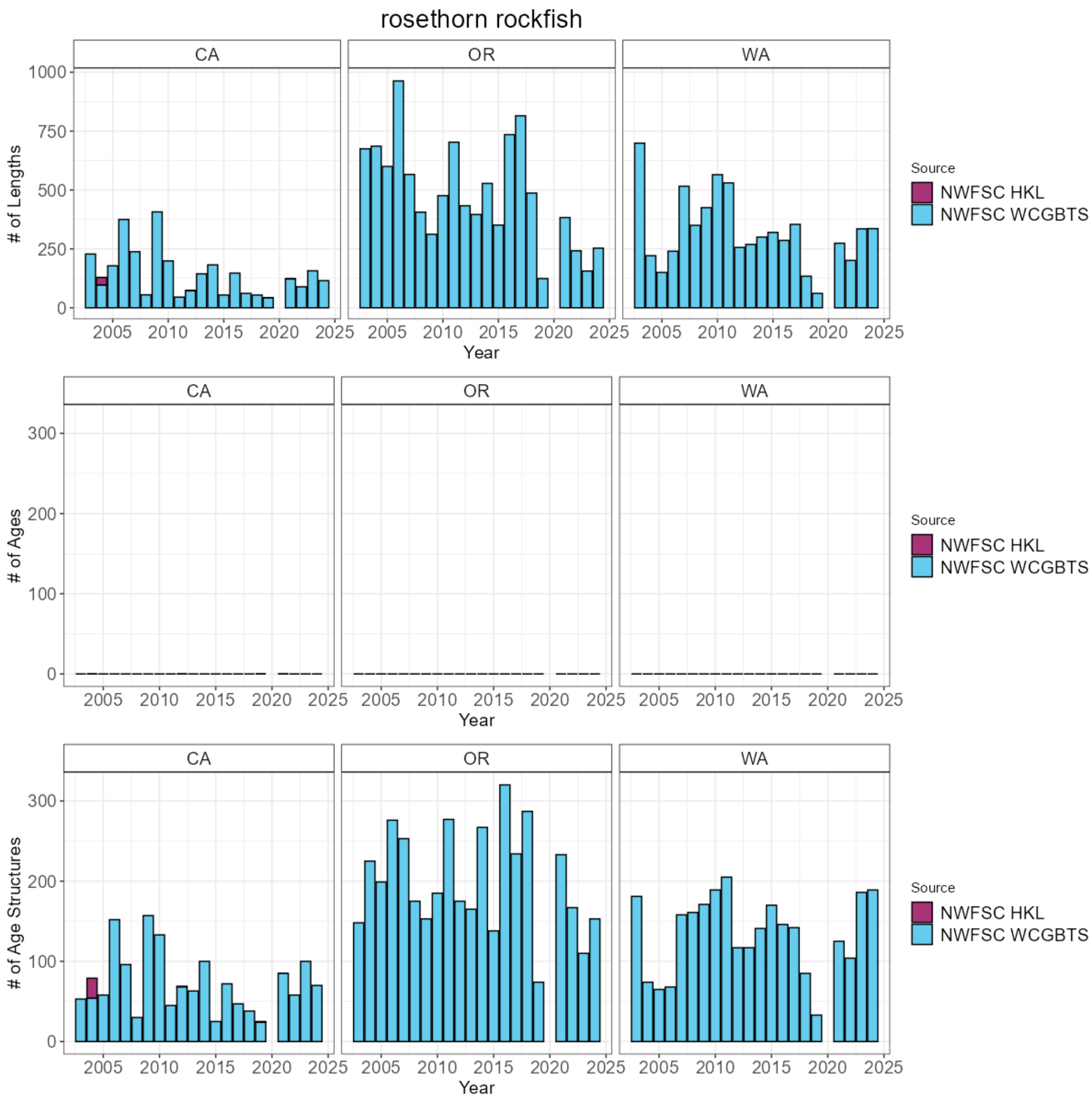

Figure 30: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.31 Rougheye and blackspotted rockfish

The most recent assessment of rougheye and blackspotted rockfish was a benchmark assessment conducted in 2025. Across available data, rougheye and blackspotted rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 27 positive tows per year. Across the coast a total of 549 maturity samples have been collected with 549 read maturities.

Table 31: Total number of available lengths, read ages, and unread age structures by data source and state between for rougheye and blackspotted rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	1	0	1
CA	NWFSC WCGBTS	15	14	0
OR	NWFSC WCGBTS	1,104	919	13
WA	NWFSC WCGBTS	1,182	1,029	3

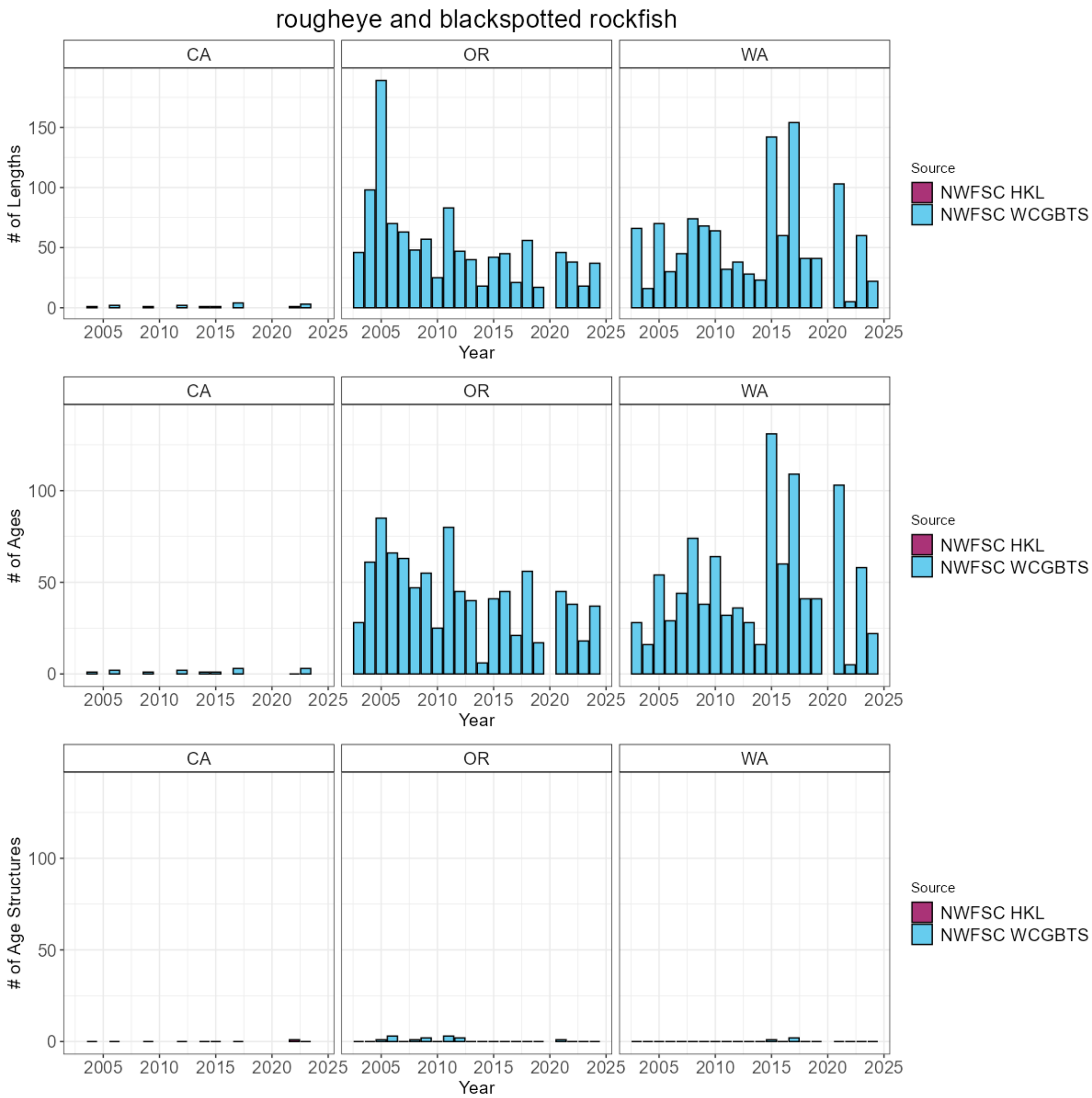

Figure 31: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.32 Sablefish

The most recent assessment of sablefish was a benchmark assessment conducted in 2025. Across available data, sablefish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 404 positive tows per year. Across the coast a total of 1312 maturity samples have been collected with 1157 read maturities.

Table 32: Total number of available lengths, read ages, and unread age structures by data source and state between for sablefish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	50,794	13,641	6,031
OR	NWFSC WCGBTS	36,899	9,952	4,228
WA	NWFSC WCGBTS	13,340	3,931	1,537

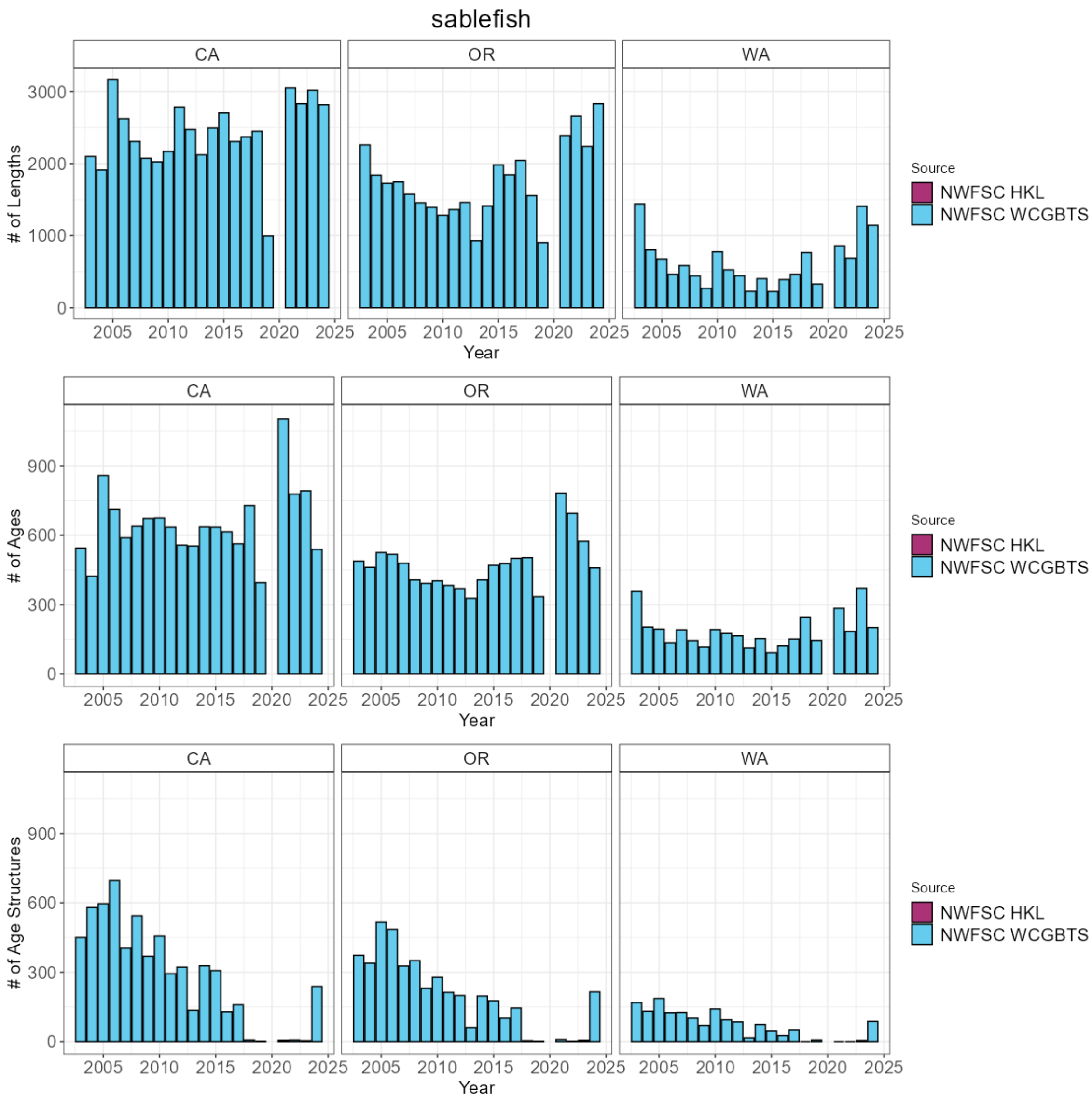

Figure 32: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.33 Sharpchin rockfish

The most recent assessment of sharpchin rockfish was a data-moderate assessment conducted in 2013. Across available data, sharpchin rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 41 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 33: Total number of available lengths, read ages, and unread age structures by data source and state between for sharpchin rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	13	0	13
CA	NWFSC WCGBTS	3,704	0	1,874
OR	NWFSC WCGBTS	10,182	0	4,153
WA	NWFSC WCGBTS	6,869	0	3,072

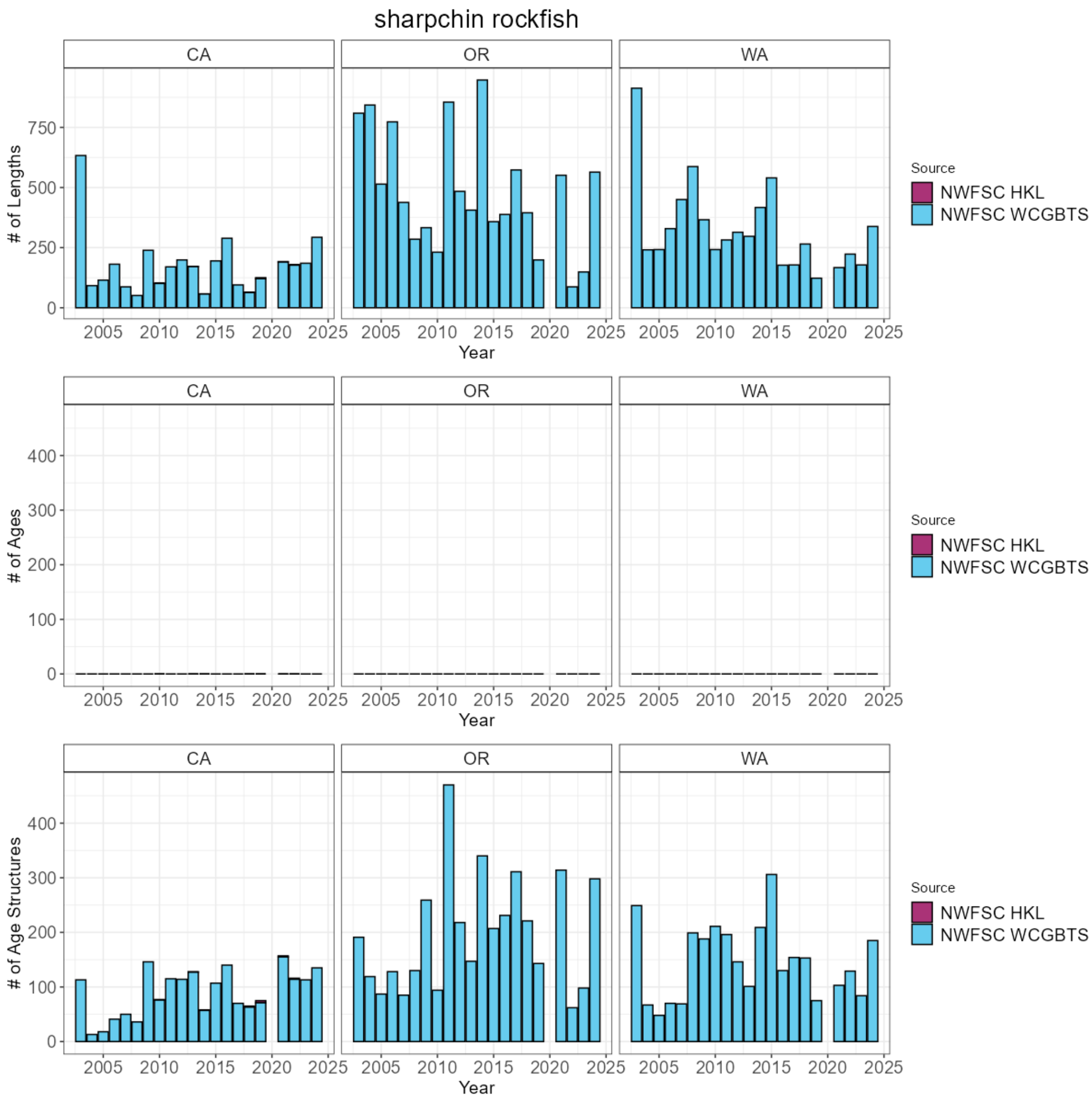

Figure 33: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.34 Shortspine thornyhead

The most recent assessment of shortspine thornyhead was a data-moderate assessment conducted in 2023. Across available data, shortspine thornyhead have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 308 positive tows per year. Across the coast a total of 1136 maturity samples have been collected with 862 read maturities.

Table 34: Total number of available lengths, read ages, and unread age structures by data source and state between for shortspine thornyhead.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	48,129	0	12,092
OR	NWFSC WCGBTS	42,573	0	7,321
WA	NWFSC WCGBTS	12,585	0	2,724

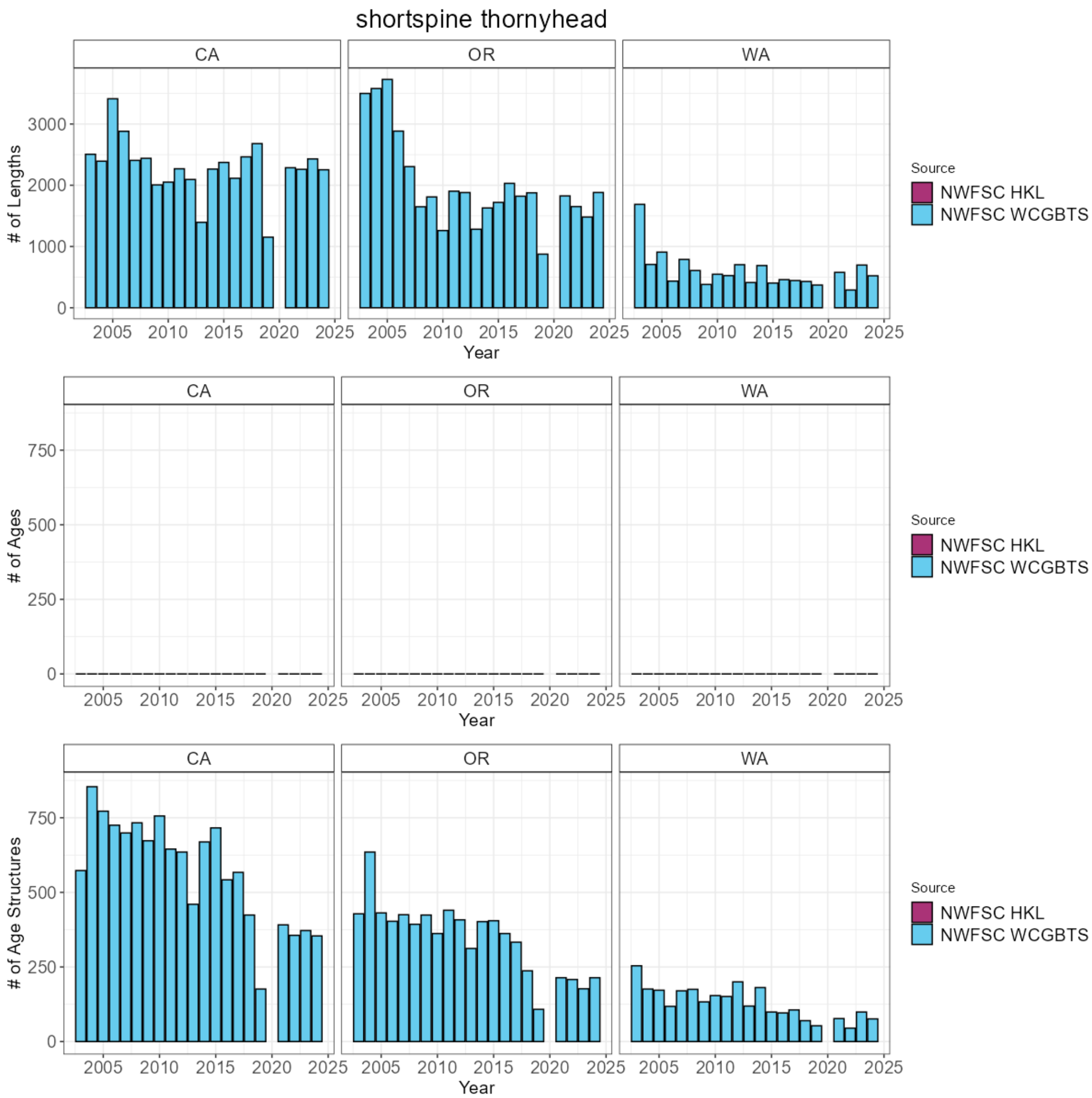

Figure 34: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.35 Silvergray rockfish

To date, no assessment or analysis has been conducted on silvergray rockfish. Across available data, silvergray rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 6 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 35: Total number of available lengths, read ages, and unread age structures by data source and state between for silvergray rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	4	0	3
CA	NWFSC WCGBTS	15	0	15
OR	NWFSC WCGBTS	481	0	393
WA	NWFSC WCGBTS	425	0	398

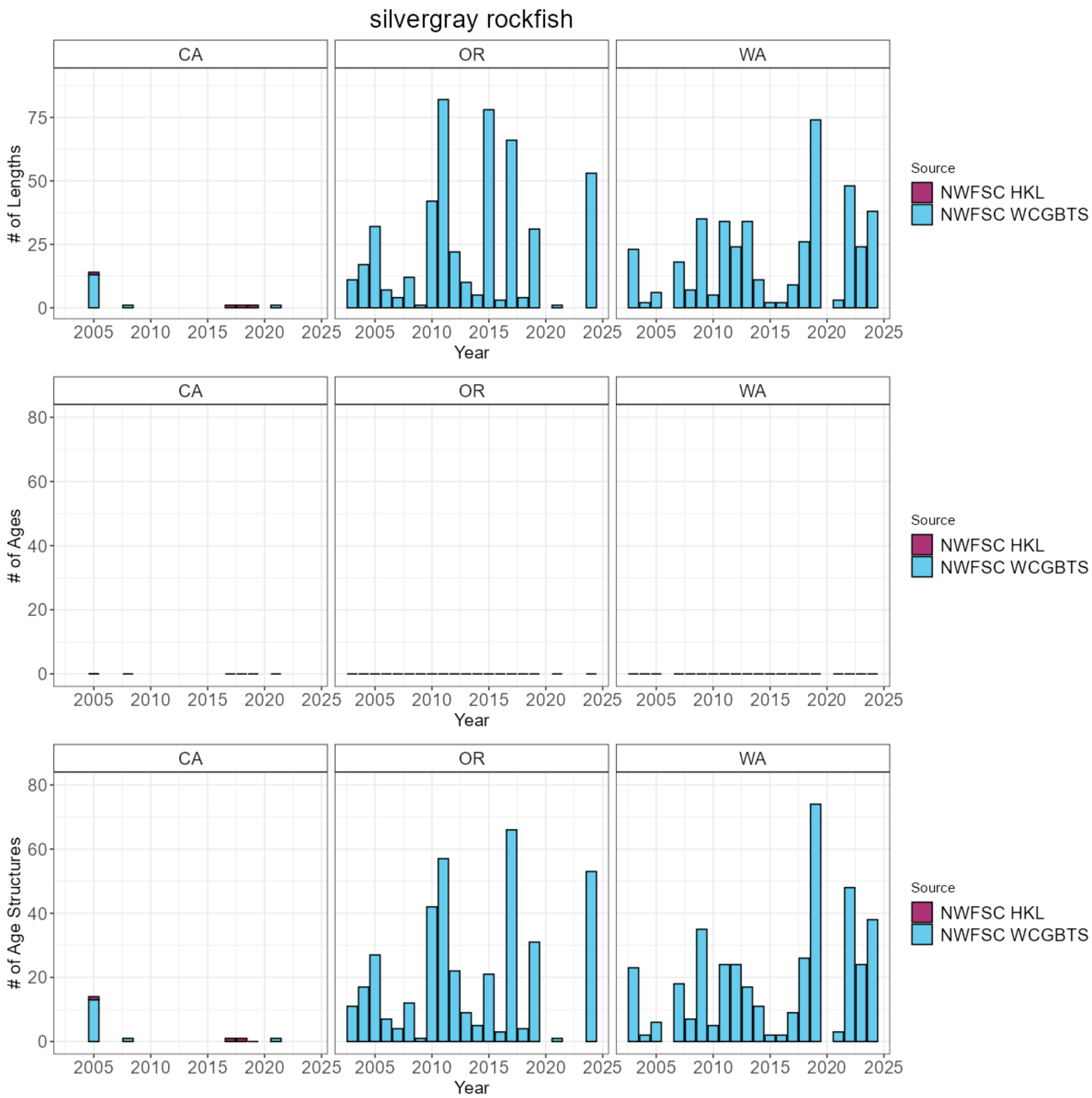

Figure 35: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.36 Splitnose rockfish

The most recent assessment of splitnose rockfish was a benchmark assessment conducted in 2009. Across available data, splitnose rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 122 positive tows per year. Across the coast a total of 0 maturity samples have been collected with 0 read maturities.

Table 36: Total number of available lengths, read ages, and unread age structures by data source and state between for splitnose rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	33,681	1,573	6,123
OR	NWFSC WCGBTS	17,858	1,044	3,584
WA	NWFSC WCGBTS	3,145	294	654

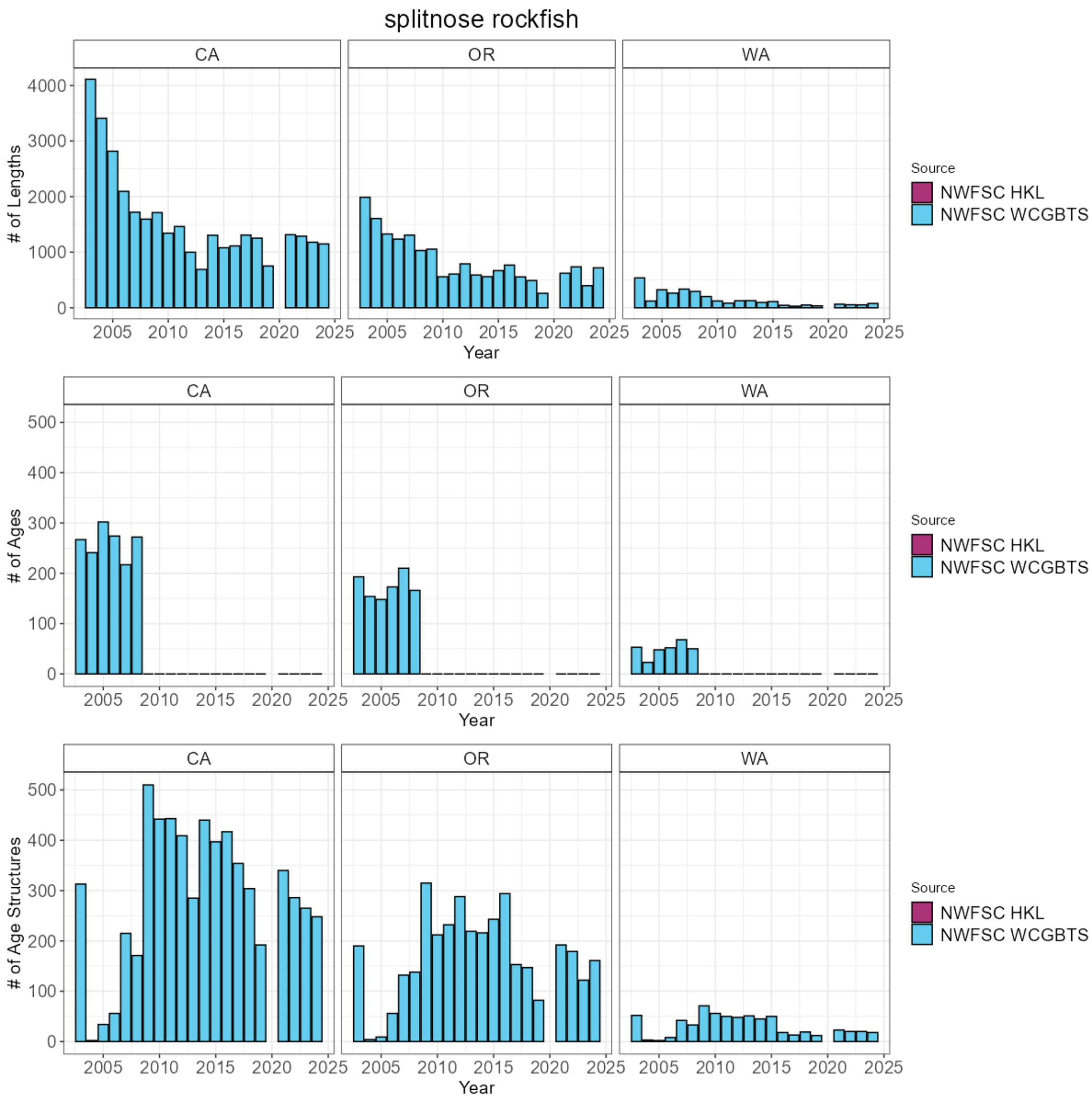

Figure 36: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.37 Squarespot rockfish

The most recent assessment of squarespot rockfish was a data-moderate assessment conducted in 2021. Across available data, squarespot rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 28 positive tows per year. The NWFSC WCGBTS survey has an average of 9 positive tows per year. Across the coast a total of 118 maturity samples have been collected with 118 read maturities.

Table 37: Total number of available lengths, read ages, and unread age structures by data source and state between for squarespot rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	2,309	344	1,862
CA	NWFSC WCGBTS	4,929	402	1,210
OR	NWFSC WCGBTS	4	1	1

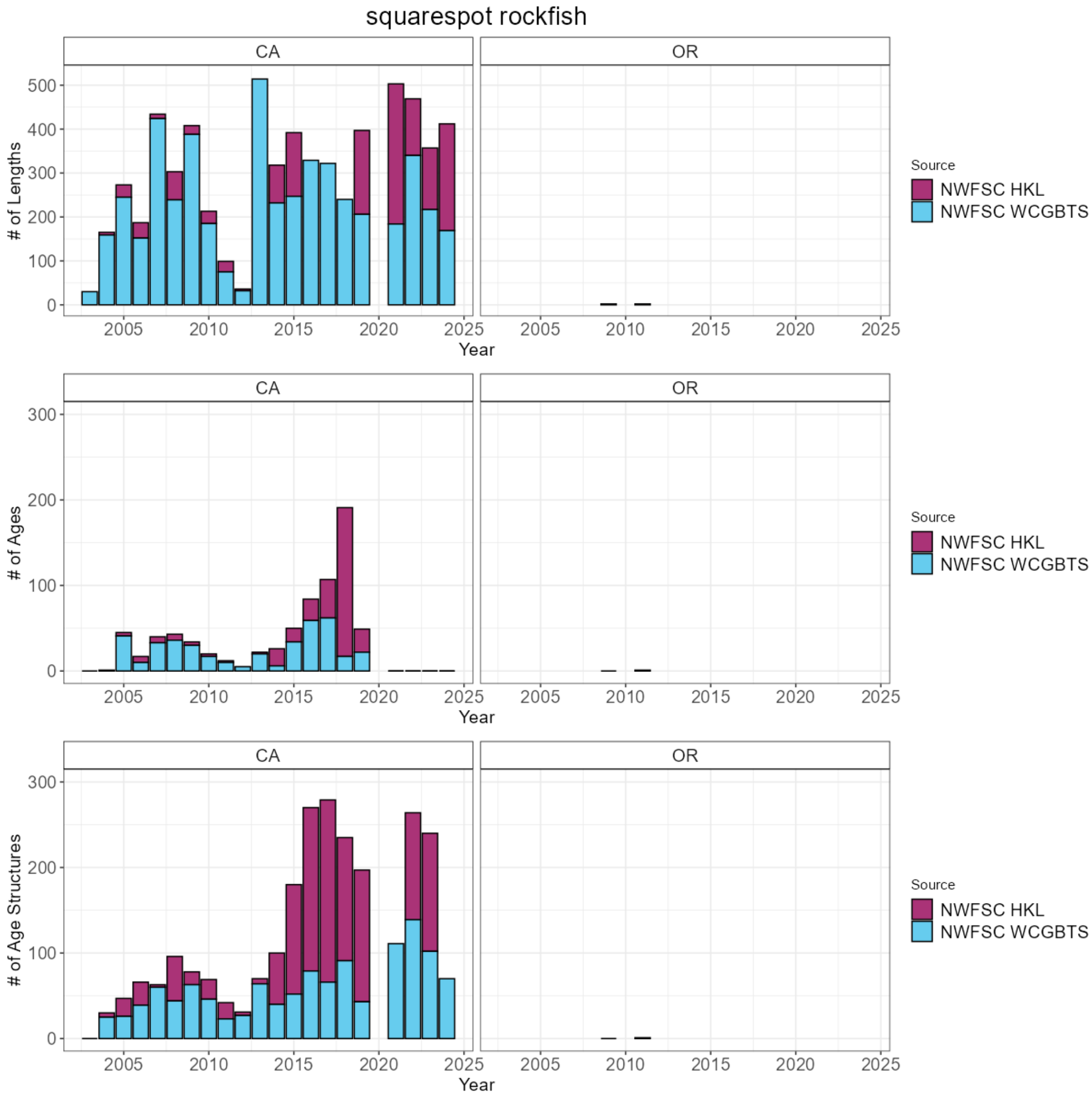

Figure 37: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.38 Starry rockfish

To date, no assessment or analysis has been conducted on starry rockfish. Across available data, starry rockfish have been observed and sampled by the NWFSC HKL survey. The NWFSC HKL survey has an average of 52 positive tows per year. Across the coast a total of 340 maturity samples have been collected with 0 read maturities.

Table 38: Total number of available lengths, read ages, and unread age structures by data source and state between for starry rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	3,692	0	3,566
CA	NWFSC WCGBTS	98	0	91

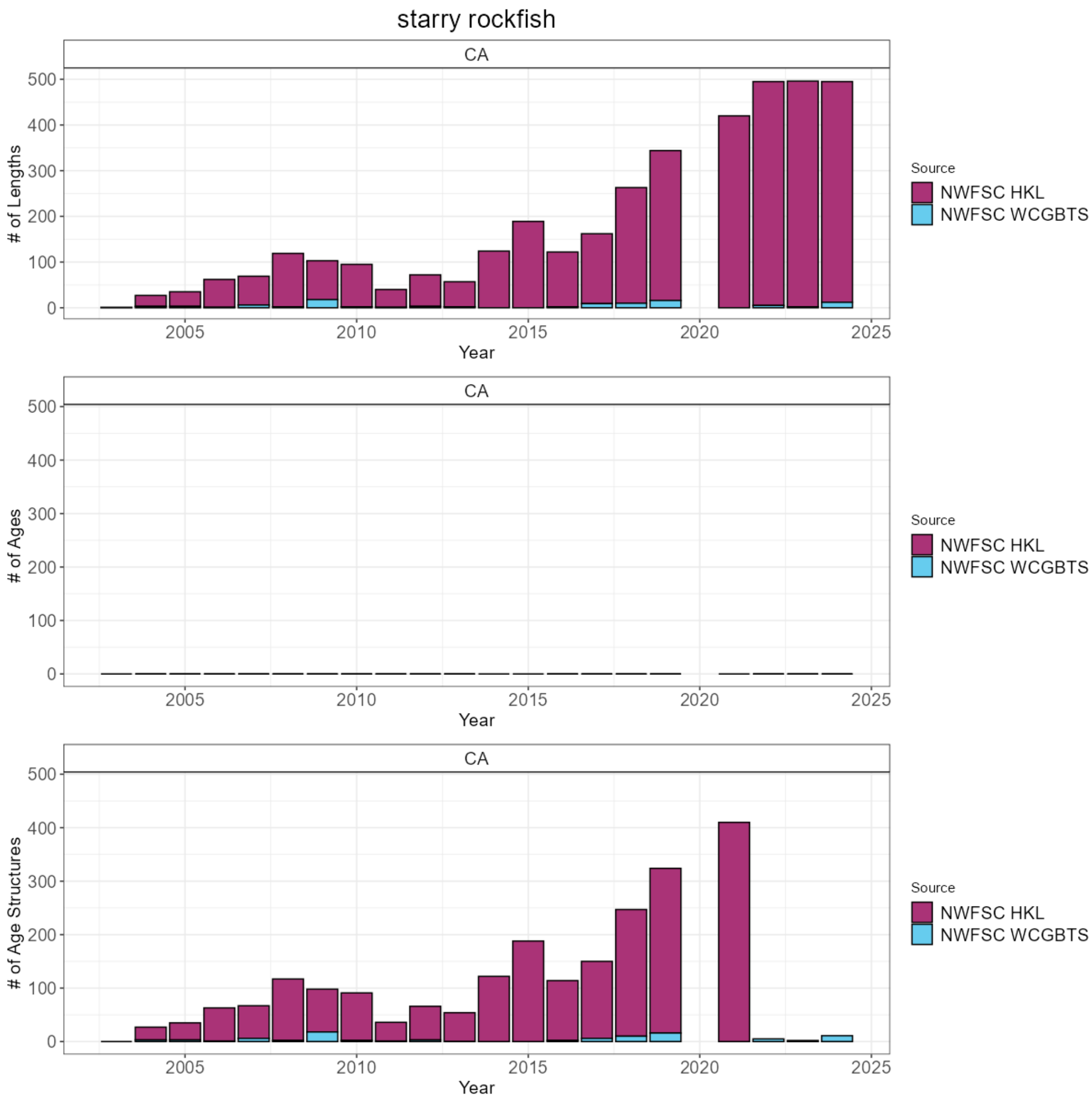

Figure 38: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.39 Stripetail rockfish

To date, no assessment or analysis has been conducted on stripetail rockfish. Across available data, stripetail rockfish have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 133 positive tows per year. Across the coast a total of 67 maturity samples have been collected with 67 read maturities.

Table 39: Total number of available lengths, read ages, and unread age structures by data source and state between for stripetail rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	3	0	3
CA	NWFSC WCGBTS	42,456	0	8,619
OR	NWFSC WCGBTS	8,886	0	2,262
WA	NWFSC WCGBTS	1,492	0	508

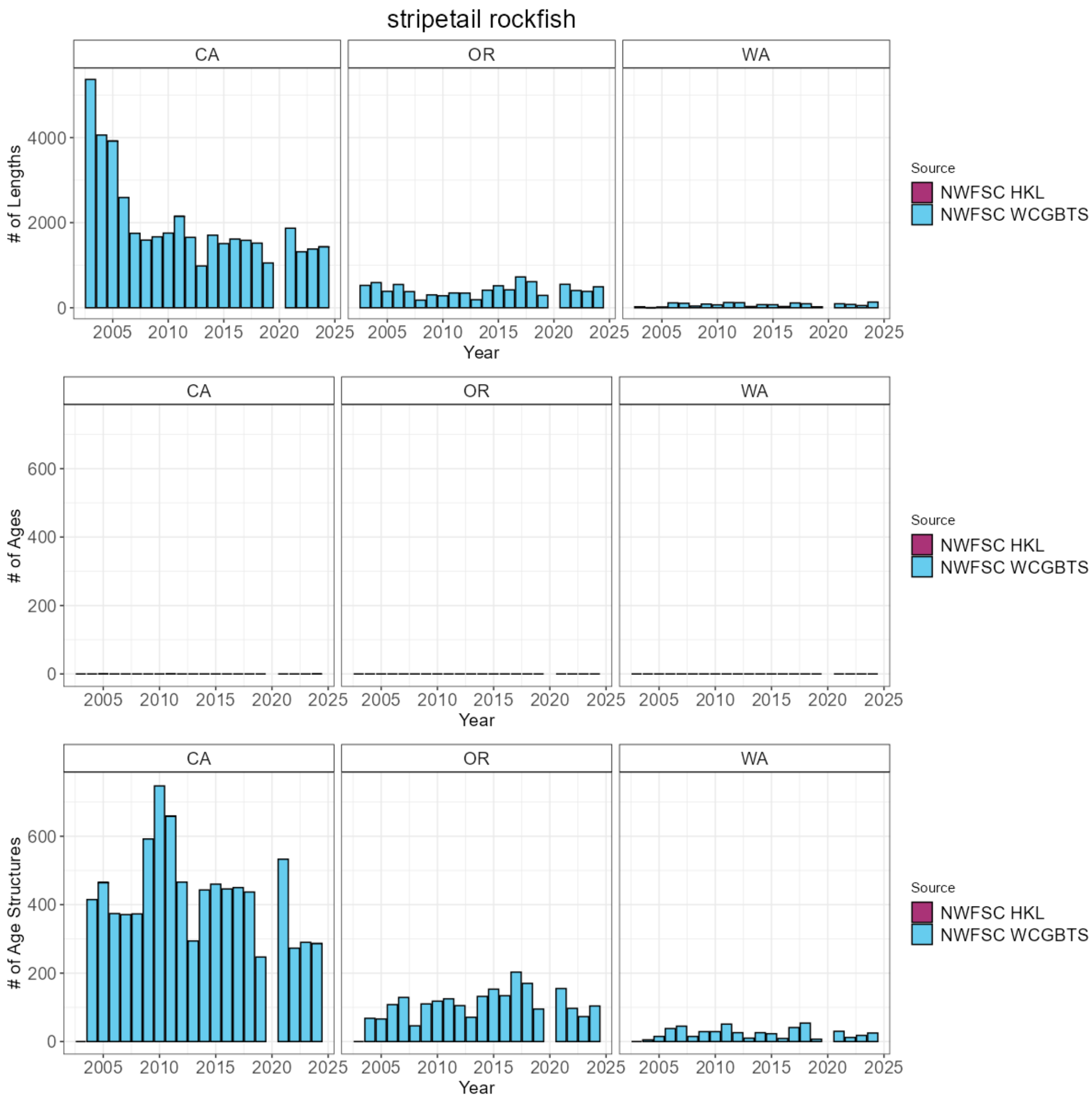

Figure 39: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.40 Vermilion and sunset rockfish

The most recent assessment of vermilion and sunset rockfish was a benchmark assessment conducted in 2021. Across available data, vermilion and sunset rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 107 positive tows per year. The NWFSC WCGBTS survey has an average of 12 positive tows per year. Across the coast a total of 1728 maturity samples have been collected with 1405 read maturities.

Table 40: Total number of available lengths, read ages, and unread age structures by data source and state between for vermilion and sunset rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	42,075	15,460	16,189
CA	NWFSC WCGBTS	3,432	1,753	717
OR	NWFSC WCGBTS	2	2	0

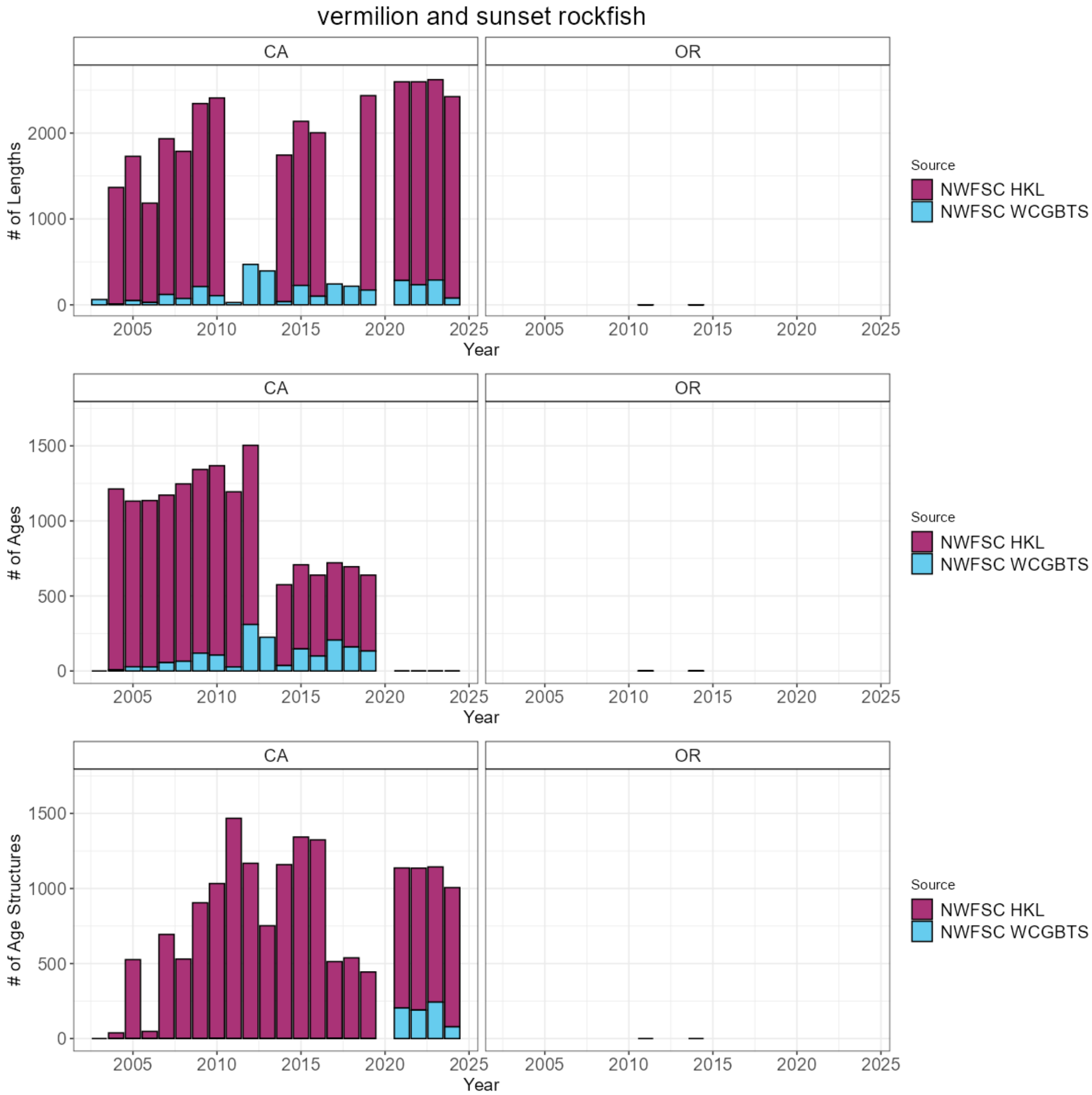

Figure 40: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.41 Widow rockfish

The most recent assessment of widow rockfish was an update assessment conducted in 2025. Across available data, widow rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 14 positive tows per year. The NWFSC WCGBTS survey has an average of 24 positive tows per year. Across the coast a total of 322 maturity samples have been collected with 92 read maturities.

Table 41: Total number of available lengths, read ages, and unread age structures by data source and state between for widow rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	899	0	877
CA	NWFSC WCGBTS	1,985	1,363	9
OR	NWFSC WCGBTS	2,161	1,417	10
WA	NWFSC WCGBTS	921	520	4

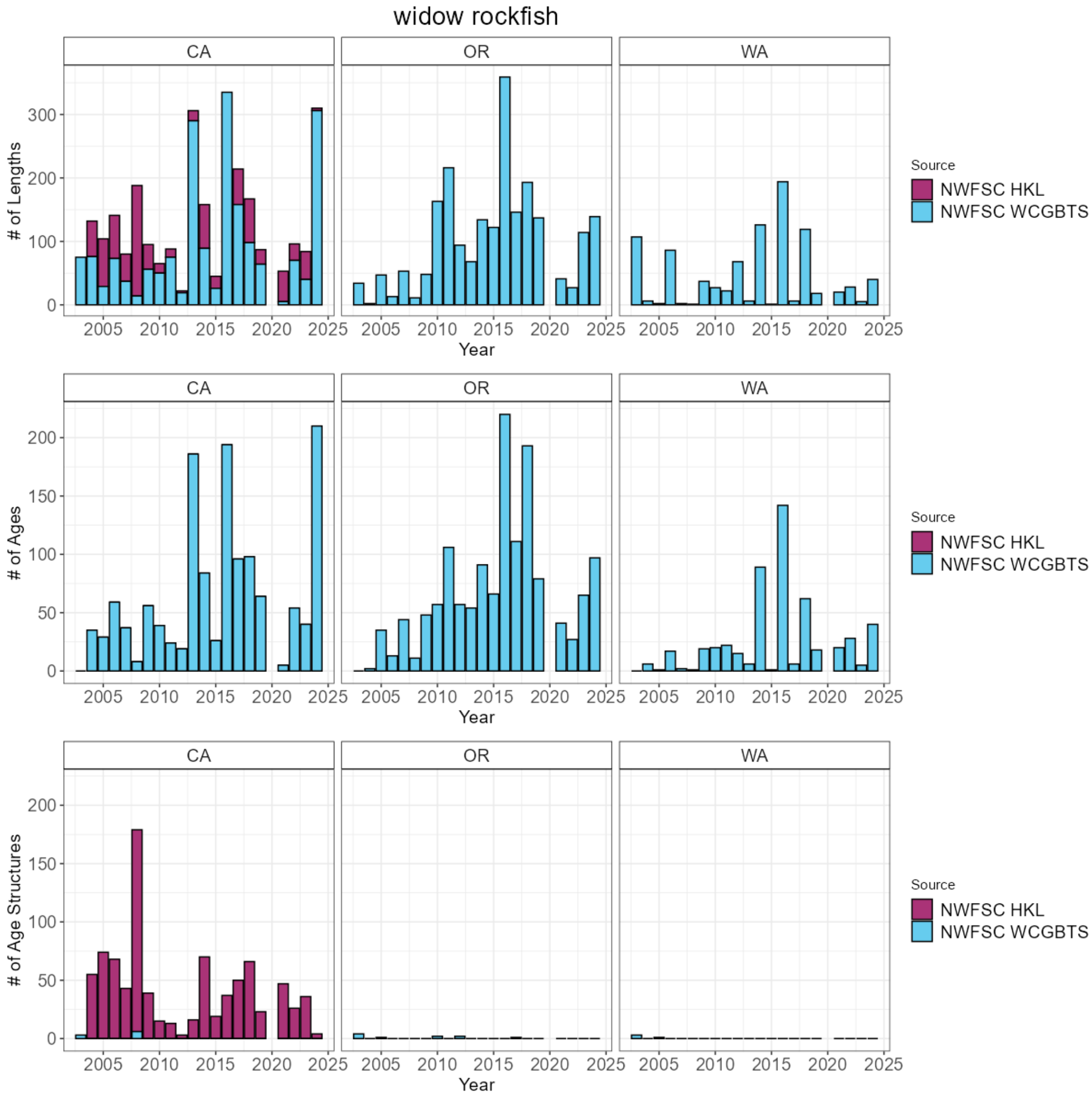

Figure 41: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.42 Yelloweye rockfish

The most recent assessment of yelloweye rockfish was an update assessment conducted in 2025. Across available data, yelloweye rockfish have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 4 positive tows per year. The NWFSC WCGBTS survey has an average of 15 positive tows per year. Across the coast a total of 708 maturity samples have been collected with 119 read maturities.

Table 42: Total number of available lengths, read ages, and unread age structures by data source and state between for yelloweye rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	159	0	151
CA	NWFSC WCGBTS	168	167	1
OR	NWFSC WCGBTS	465	457	7
WA	NWFSC WCGBTS	416	410	6

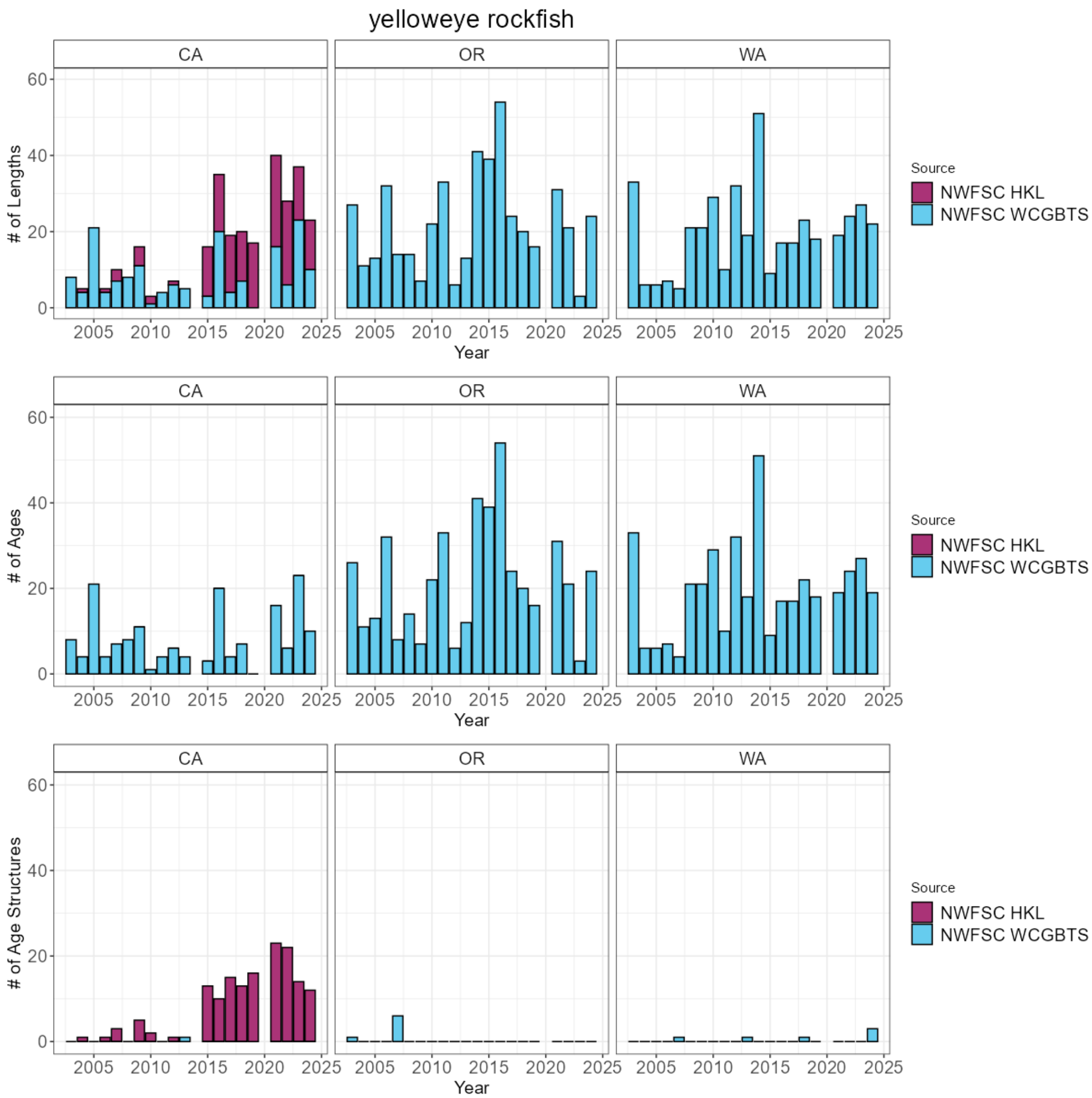

Figure 42: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.43 Yellowmouth rockfish

To date, no assessment or analysis has been conducted on yellowmouth rockfish. Across available data, yellowmouth rockfish have been observed and sampled by the NWFSC WCGBT survey. Across the coast a total of 12 maturity samples have been collected with 0 read maturities.

Table 43: Total number of available lengths, read ages, and unread age structures by data source and state between for yellowmouth rockfish.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	1	0	1
OR	NWFSC WCGBTS	572	0	303
WA	NWFSC WCGBTS	52	0	52

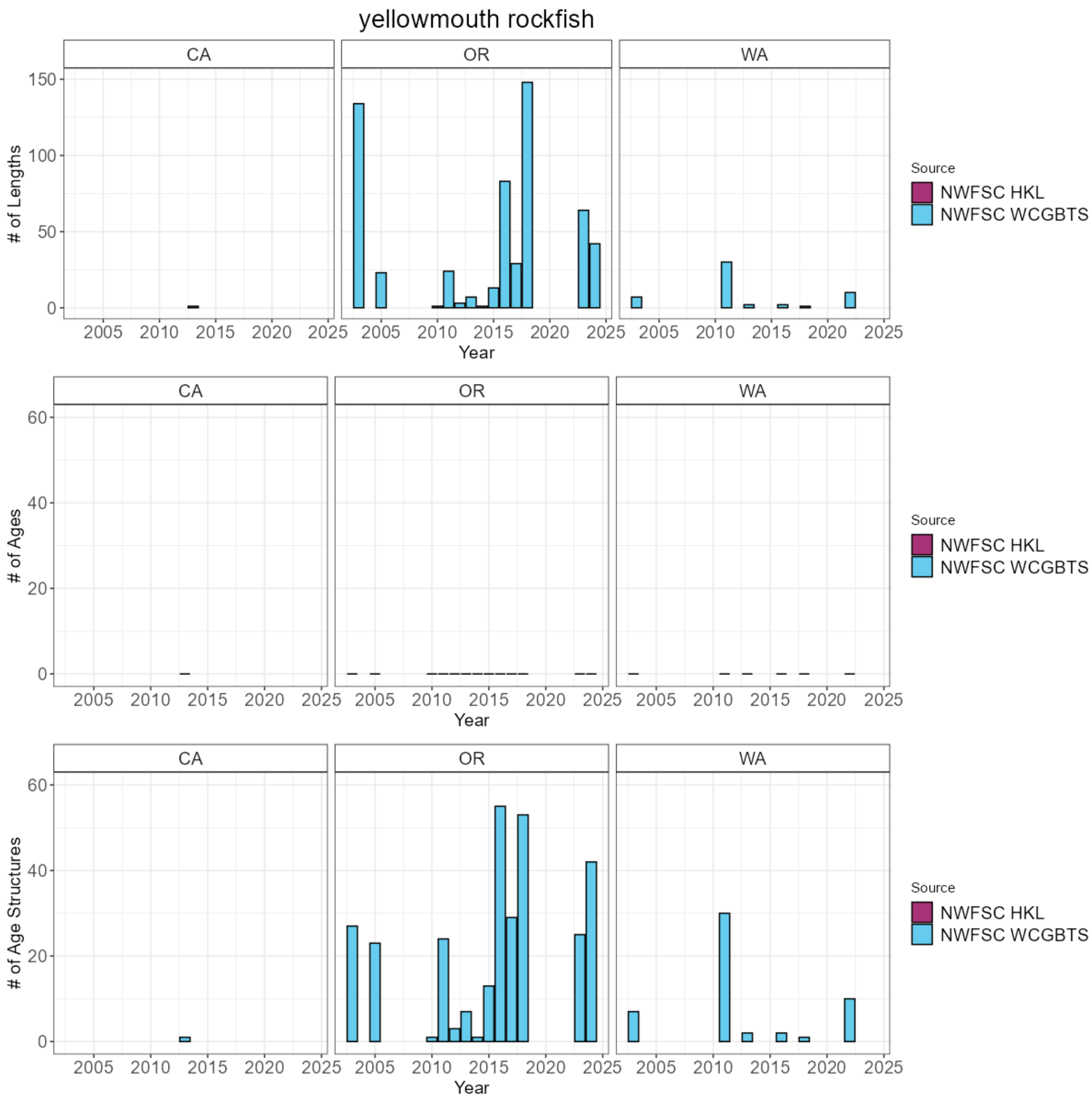

Figure 43: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.44 Yellowtail rockfish north

The most recent assessment of yellowtail rockfish north was a benchmark assessment conducted in 2025. Across available data, yellowtail rockfish north have been observed and sampled by the NWFSC WCGBT survey. The NWFSC WCGBTS survey has an average of 41 positive tows per year. Across the coast a total of 739 maturity samples have been collected with 672 read maturities. ODFW scientists are planning a research cruise to Cobb Sea Mount in the summer of 2026 to sample age and sex distributions of rockfish species.

Table 44: Total number of available lengths, read ages, and unread age structures by data source and state between for yellowtail rockfish north.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC WCGBTS	792	292	27
OR	NWFSC WCGBTS	3,912	1,947	404
WA	NWFSC WCGBTS	12,624	5,633	877

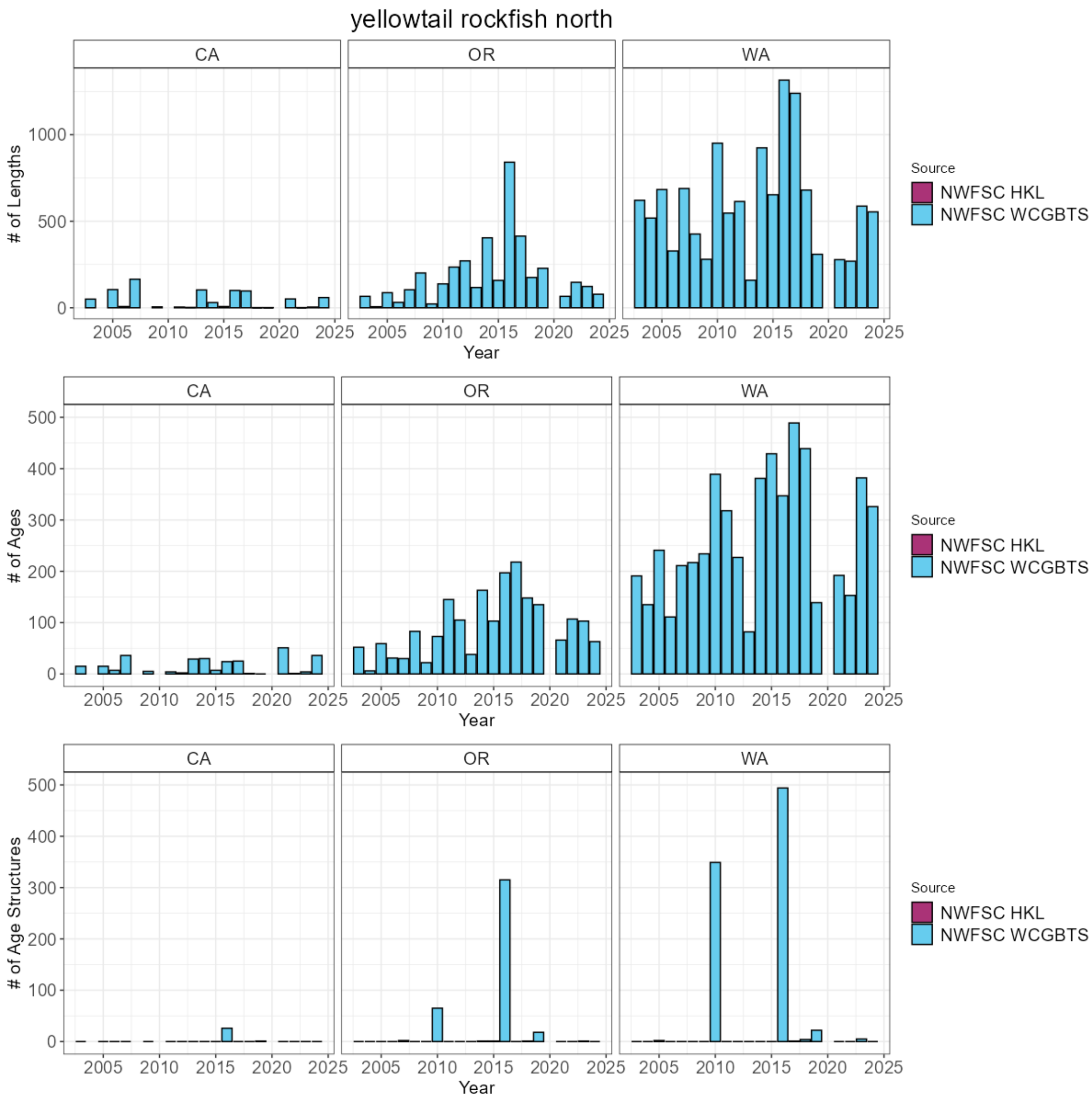

Figure 44: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

2.45 Yellowtail rockfish south

To date, no assessment or analysis has been conducted on yellowtail rockfish south. Across available data, yellowtail rockfish south have been observed and sampled by both the NWFSC WCGBT and HKL surveys. The NWFSC HKL survey has an average of 13 positive tows per year. Across the coast a total of 739 maturity samples have been collected with 672 read maturities.

Table 45: Total number of available lengths, read ages, and unread age structures by data source and state between for yellowtail rockfish south.

State	Source	Lengths	Ages	Age Structures
CA	NWFSC HKL	1,907	124	1,564
CA	NWFSC WCGBTS	1,072	534	21

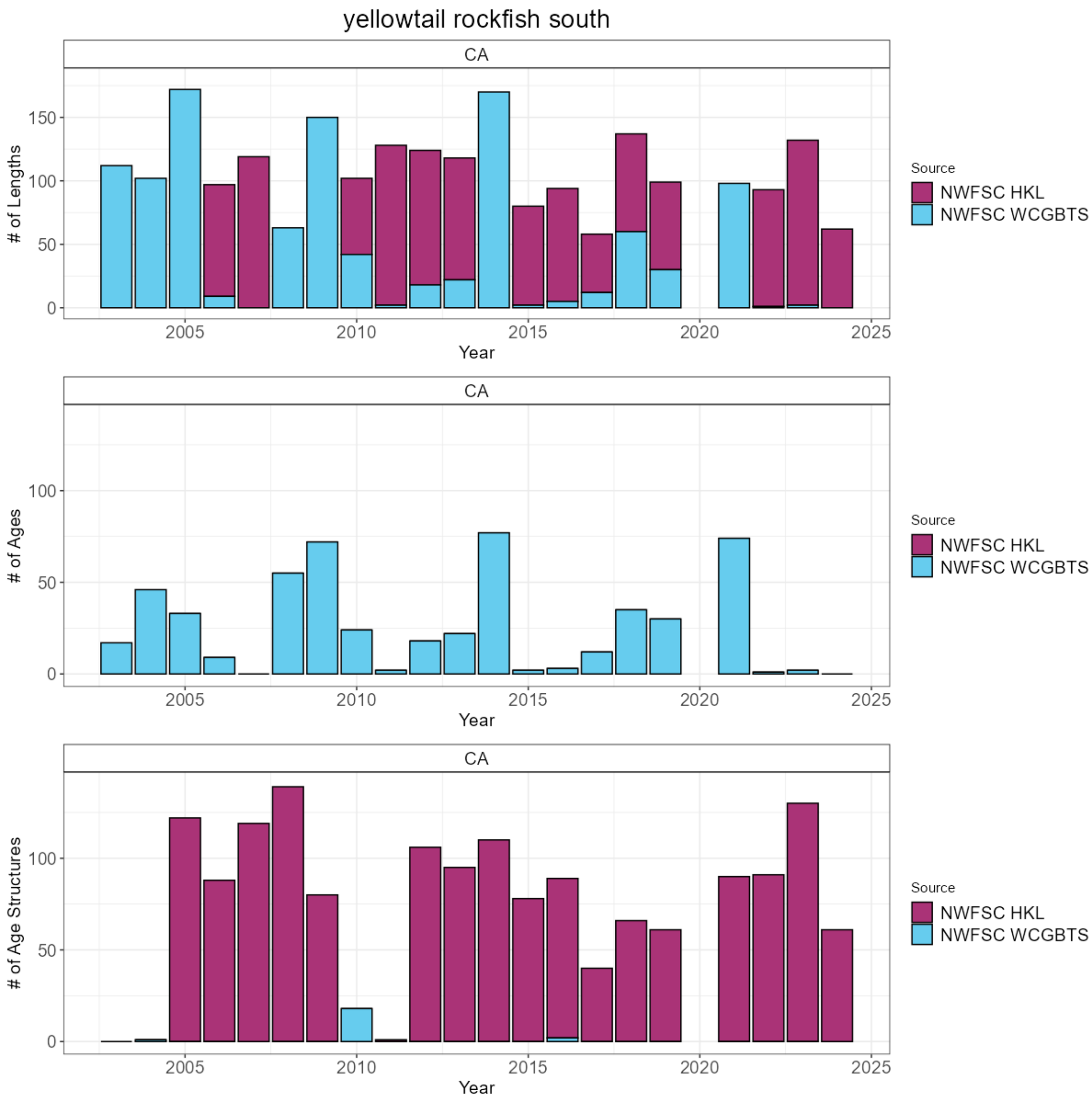

Figure 45: Total number of available lengths, read ages, and unread age structures by data source by year. Note the y-axis is unique for the number of lengths plot row compared to the number of age and age structure plot rows.

3 Index Estimation for the NWFSC WCGBT Survey

Table 46: Setting used for estimating indices of abundance for the NWFSC WCGBT survey by species.

species	family	formula	min latitude	max latitude	spatiotemporal1	spatiotemporal2
canary rockfish	delta_lognormal()	catch_weight ~ 0 + fyear + pass_scaled	31.9	49.0	iid	off
chilipepper	delta_lognormal()	catch_weight ~ 0 + fyear + pass_scaled + depth_scaled + depth_scaled_squared	31.9	46.2	iid	off
rougheye rockfish	delta_lognormal()	catch_weight ~ 0 + fyear + pass_scaled	42.0	49.0	off	off
sablefish	delta_gamma()	catch_weight ~ 0 + fyear + pass_scaled	31.9	49.0	iid	iid
widow rockfish	delta_lognormal()	catch_weight ~ 0 + fyear + pass_scaled	33.5	49.0	iid	off
yelloweye rockfish	delta_lognormal()	catch_weight ~ 0 + fyear + pass_scaled	42.0	49.0	off	off
yellowtail rockfish	delta_lognormal()	catch_weight ~ 0 + fyear*split_mendocino + pass_scaled	33.5	49.0	iid	off

4 Acknowledgement

We would like to thank the NWFSC Fisheries Resource Analysis and Monitoring survey team and their dedication to collecting high quality data that are essential to the assessment of West Coast groundfish stocks.